Shared SDG Labels Don't Mean Shared Framing — Embedding-Based Register-Topic Decomposition of Research-Policy Divergence Across the SDGs

Contents

1 Introduction **4**

2 Literature Review **5**

2.1 AI and the SDGs: What Bibliometrics Shows 5
2.2 A Three-Level Framework of Research-Policy Alignment 6
2.3 Research-Policy Alignment in AI and Sustainability 7
2.4 Semantic Methods: Rationale and Cautions 8
2.5 Research Gap 9
2.6 Research Questions and Hypotheses 10

3 Methodology **10**

3.1 Research Corpus 10
3.2 Policy Corpus 11
3.3 Token-Aware Segmentation 12
3.4 Embedding Model and Normalisation 12
3.5 Supervised Reference Classifier 13
3.6 Coverage Gap Analysis 14
3.7 Semantic Gap Analysis 14
3.8 Register Adjustment via Iterative Nullspace Projection (INLP) 15
3.9 Coverage–Semantic Interaction 17

4 Results **18**

4.1 Supervised Reference Classifier and Coverage Gap 19
4.2 Semantic Gap after Register Removal 21

4.3 Coverage–Semantic Interaction: Results . . . 23
4.4 Robustness of the Gap Rankings . . . 25

5 Discussion **27**
5.1 The cancellation is real, not artefactual . . . 27
5.2 Two communities, two dimensions . . . 27
5.3 The register decomposition: what the raw gap actually measures . . . 29
5.4 Implications: topical mention is not alignment . . . 29
5.5 Limitations . . . 30
5.6 The framework as the main contribution . . . 31

6 Conclusion **31**

A Retrieval and Query Chain **39**

B Diagnostics for Reference Classifier **40**
B.1 SDG 4 Lexical Artefact Audit: The ML 'Learning' Overlap . . . 40
B.2 Centroid Similarity . . . 41

C Lexical Illustration of the Semantic Gap **43**

D Distance-Functional Robustness Table **45**

E Sample-Stability Robustness Check **48**

F Model Selection: Grid Search Protocol and Rationale **49**

G Register Removal: Convergence and Validation **51**
G.1 Convergence . . . 51
G.2 Construct Validation . . . 53

H Supplementary Cross-Method Data **57**
H.1 Cross-Method Coverage and Semantic Gap Values . . . 57
H.2 Rank Robustness: Cross-Sensitivity and Encoder Sensitivity . . . 58
H.3 Encoder Sensitivity . . . 58
H.4 Assignment-Method Comparison: Supervised vs. Nearest-Centroid . . . 58
H.5 Raw-Value Correlation: Metric-Sensitivity Companion . . . 59
H.6 Detailed Policy Profiles by Source Family . . . 59

I Pooled Regression: Coverage Predictors and the Semantic Gap **72**

J Declaration of AI Use **73**

Abstract

Shared SDG labels do not establish that research and policy *frame* a goal similarly. This paper contributes a measurement framework that separates research-policy divergence into two independent dimensions: coverage (which SDGs each corpus prioritises) and within-SDG semantic framing (how differently the same SDG is discussed); framing is itself a composite signal of topic (substantive content) and register (communicative conventions), and this paper isolates topic from register using Iterative Nullspace Projection (INLP). Using frozen Sentence-BERT embeddings and a supervised logistic-regression classifier (LR) trained on 52,835 SDG-labelled reference segments (test macro-$F_1 = 0.816$), with multi-layer perceptron (MLP) and zero-shot nearest-centroid variants, this study scores 3,105,144 segments from 2,536,771 AI-for-sustainability abstracts and 40,597 segments from 6,367 institutional policy sources, and measures both dimensions across all 17 SDGs.

The research-side coverage ranking reproduces the leading position of SDG 3 established by prior keyword- and citation-based inventories, while the policy-side ranking (SDGs 13, 16, and 17) is established here for the first time.

The raw coverage–semantic-gap correlation is nevertheless null ($\rho = -0.012$, permutation $p = 0.959$). Decomposing within-SDG distance into topic and register components, however, reveals a systematic cancellation: the adjusted semantic gap is positively associated with coverage ($\rho = 0.544$, $p = 0.025$) while the removed register component is negatively associated ($\rho = -0.390$, $p = 0.120$). All p-values are two-sided Monte Carlo permutation p-values (100,000 resamples). The linkage operates asymmetrically: policy coverage is the strongest directional predictor (positive in 9/9 configurations; $\rho = +0.589$ for MPNet LR), while research coverage shows no consistent signal. Across the full $n = 17$ SDG framework, only the adjusted association reaches significance; the register component is directionally negative but not significant, so the null raw gap reflects a partial, not complete, cancellation.

The result is nonetheless stable: the adjusted semantic-gap ranking survives every sensitivity check (encoders, classifier family, retrieval strategy, segment cap, and policy-source family) and distribution-aware metrics confirm it is a mean-direction effect. Coverage and framing should be measured and reported as separate axes of divergence; shared labels are insufficient evidence of shared framing.

Keywords: research-policy alignment; Sustainable Development Goals; semantic similarity; transformer embeddings; measurement protocol; bibliometrics.

Data and code availability. The analysis code and the frozen, processed outputs are version-controlled at `https://github.com/qmanhbeo/dissertation-bham`.

1 Introduction

Research-policy alignment is often inferred from whether scientific and policy texts address the same topics (Bornmann, Haunschild, and Marx, 2016), yet shared coverage does not establish comparable framing: a paper and a policy document can both invoke SDG 13 (Climate Action) or SDG 3 (Good Health) while emphasising different problems, actors, and interventions. Legislative agendas, ESG disclosures, and funding decisions increasingly treat SDG tags as evidence of alignment (Heilinger, Kempt, and Nagel, 2024; Heras-Saizarbitoria, Urbieta, and Boiral, 2022); this assumption is empirically unfounded.

Existing methods for mapping research to the Sustainable Development Goals (SDGs) operate mainly at the level of lexical or bibliometric correspondence: keyword dictionaries, Boolean queries, or citation and mention counts (Armitage, Lorenz, and Mikki, 2020; Kashnitsky et al., 2024). They measure whether a text is *about* a given SDG, not how it frames it. The research-side coverage distribution reproduces the leading position of SDG 3 that those methods established, while the policy-side dominance of SDGs 13, 16, and 17 is established here, anchoring external validity before the paper turns to framing.

This paper proposes that research-policy alignment has at least two separable dimensions: coverage divergence (which SDGs each corpus prioritises) and semantic framing divergence (how differently they discuss the same SDG). Its contribution is a measurement framework. This framework advances AI–SDG mapping from Level 1 (lexical correspondence) to Level 2 (observed textual-semantic proximity in embedding space; Section 2.2), treating semantic proximity as a register-sensitive proxy for framing, not evidence of substantive alignment. The framework operationalises both dimensions across all 17 SDGs with three frozen sentence-transformer encoders (MPNet, MiniLM, SciBERT; Section 3.4) and three classifier variants – logistic regression, multi-layer perceptron, and zero-shot nearest-centroid – stress-tested against encoders, classifier family, retrieval strategy, segment cap, and policy-source family choices. A novel SDG-stratified INLP adaptation separates register from topic.

The near-zero raw coverage-vs-semantic-gap correlation may reflect cancellation: topic divergence rises with coverage divergence while register divergence falls with it, and the two appear to offset (permutation $p = 0.025$ and $p = 0.120$ respectively, $n = 17$ SDGs); the directional decomposition further suggests this linkage operates primarily through policy coverage, not research coverage. This matters because SDG tagging is now institutionalised; if shared labels mask a register–topic cancellation, that threatens how the SDG-monitoring ecosystem reads its own signals.

The measurement choice changes the answer at the extremes: the raw gap (naive baseline) makes SDG 3 the most divergent goal, while the adjusted semantic gap after register removal identifies SDG 17 instead, which is also the goal most sensitive to how divergence is measured. SDG 15 is the stably least-divergent goal under both.

The central research question is: *To what extent do academic AI-for-sustainability research and international institutional SDG policy discourse differ in SDG coverage and semantic framing across the 17 SDGs, and how are these two dimensions related?*

2 Literature Review

2.1 AI and the SDGs: What Bibliometrics Shows

Existing assessments of AI and the SDGs rest on expert views or publication counts rather than on the texts themselves (Vinuesa et al., 2020). Broader SDG scientometrics and the AI-specific inventories converge on one robust finding: SDG 3 (Good Health) and SDG 7 (Affordable and Clean Energy) are the most addressed goals in the AI-specific studies, followed by SDG 4 (Quality Education), SDG 13 (Climate Action), SDG 11 (Sustainable Cities and Communities), and SDG 16 (Peace, Justice and Strong Institutions). The evidence base spans a citation-based corpus of roughly 25,000 publications on the Millennium and Sustainable Development Goals (Bautista-Puig et al., 2021), 20,511 AI publications (2001–2020) (Singh et al., 2024), 1,288 big-data and AI publications (Nedungadi et al., 2024), 792 AI-and-SDG research articles (Gohr et al., 2025), and 108 AI-for-social-good projects (Cowls et al., 2021). None of these studies measures within-SDG semantic framing.

SDG publication mapping is methodologically contingent. Armitage, Lorenz, and Mikki (2020) shows that what counts as SDG-relevant depends on interpretive choices about queries, labels, and databases; Kashnitsky et al. (2024) reach the same caution from a broader comparison of Boolean, machine-learning, and hybrid mapping systems, finding no single best approach and no single "golden" validation set. The dependence persists in AI-based mapping: keyword baselines still omit publications that never state an SDG keyword explicitly, so the reported corpus boundary remains query-dependent (Yin et al., 2025). The embedding-based approach used here reduces some vocabulary-level brittleness of keyword dictionaries, but its results remain conditional on corpus construction, centroid provenance, and model choice.

The SDGs themselves are interdependent by design, yet no operational framework specifies how (Nilsson, D. Griggs, and Visbeck, 2016); single-SDG assignment is therefore a simplifying assumption. Le Blanc (2015), in a network analysis of the proposed SDGs, treats SDG 17 (Partnerships for the Goals) as the dedicated goal for cross-cutting means of implementation: finance, trade, technology transfer, and capacity building, excluding it from cross-goal mapping because its targets are implementation infrastructure rather than substantive domains. The D. J. Griggs et al. (2017) guide likewise describes SDG 17 as the main enabler of the whole framework. Unlike every other SDG, whose discourse centres on a shared substantive topic, SDG 17 policy discourse concerns coordinating and resourcing implementation itself, a conceptual space that technical AI research abstracts are not built to occupy.

2.2 A Three-Level Framework of Research-Policy Alignment

This paper proposes that research-policy alignment be understood as a three-level construct rather than a binary property of shared labels:

Level 1: lexical or bibliometric correspondence. The dominant mode of existing SDG-mapping work: assigning texts to goals through keyword dictionaries, Boolean queries, or citation and mention counts (Armitage, Lorenz, and Mikki, 2020; Kashnitsky et al., 2024). Necessary but weak: shared vocabulary does not guarantee shared meaning (Bornmann, Haunschild, and Marx, 2016), and keyword methods are brittle to vocabulary and database choices (Armitage, Lorenz, and Mikki, 2020).

Level 2: observed textual-semantic proximity in embedding space. Sentence embeddings locate texts in a shared semantic space where proximity reflects distributional patterns of co-occurrence (Harris, 1954; Reimers and Gurevych, 2019), and encoder-based SDG association has been benchmarked as a strong baseline (Gjorgjevikj et al., 2025). This level detects divergence that keyword methods miss. It also makes divergence structurally visible: observed proximity decomposes into separable components (coverage versus framing, and topic versus register within framing), and because the decomposition operates on the texts themselves, its findings are reproducible and falsifiable rather than judgement-dependent (Section 3.8). The present study is positioned at this level, and the limits of that measurement are developed in Section 2.4 below.

Level 3: substantive comparison of implied problems, actors, and interventions. Whether the two discourses address the same problems, involve the same actors, and propose the same interventions (Cash et al., 2003; Guston, 2001; Haas, 1992). Because this requires interpreting causal beliefs, legitimacy, and institutional mechanism (Caplan, 1979; Haas, 1992), it lies beyond unsupervised text analysis and is left to qualitative follow-up.

Advancing from Level 1 to Level 2 is the study's concrete methodological contribution; Level 3 names the boundary it deliberately does not cross, and its interpretative questions are complementary rather than prerequisite to that contribution. Figure 1 illustrates this decomposition.

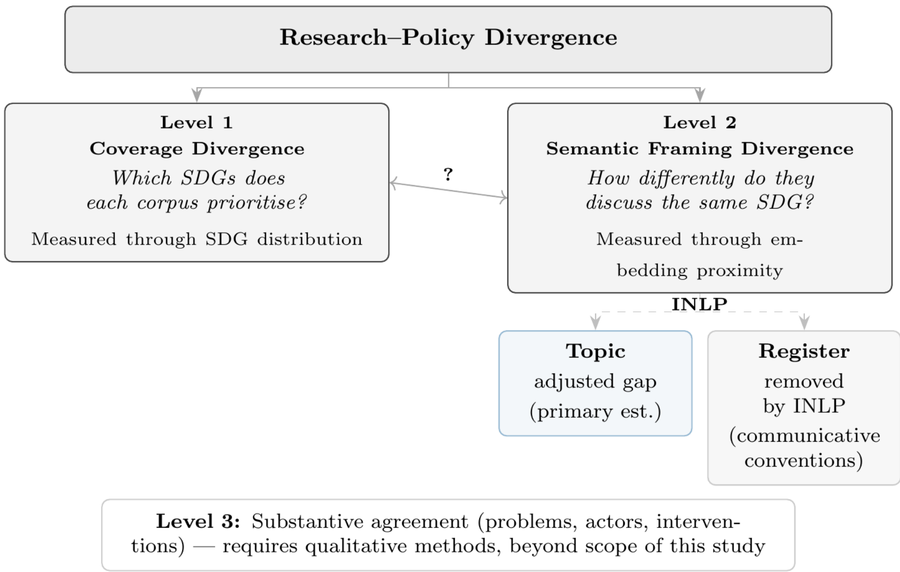

Figure 1. Conceptual framework: research–policy divergence separated into coverage and semantic framing dimensions, positioned within the three-level alignment construct.

2.3 Research-Policy Alignment in AI and Sustainability

A growing literature documents structural tensions between what AI research produces and what sustainability policy requires: corporate AI research concentrates on pre-deployment technical problems, creating a gap between research priorities and the deployment impacts governance must address (Strauss et al., 2025); global AI metrics emphasise technical and economic performance while policy discourse stresses ethical and social dimensions aligned with SDG concerns (Sioumalas-Christodoulou and Tympas, 2025); and research and policy communities can share trustworthy-AI terminology while differing in term salience and focal concerns (Toney-Wails, Curlee, and Probasco, 2024). The pattern extends beyond AI: in conservation science, 59% of policy-prioritised research topics appear in fewer than 15 papers (McCarthy et al., 2026).

Adjacent scientometric work measures research-policy connection through policy-document mentions rather than semantic framing: only 1.2% of 191,276 climate-change publications have at least one policy mention (Bornmann, Haunschild, and Marx, 2016).

Theory explains why such divergence is the default rather than a failure. Knowledge systems support action when they produce information that is simultaneously salient, credible, and legitimate (Cash et al., 2003); because these are properties of how information is framed and packaged rather than of which topics it covers, they motivate measuring framing alongside coverage, not only topical presence. The epistemic-communities framework explains why shared framing matters for policy coordination under uncertainty: expert communities influence policy by articulating cause–effect relationships and framing issues for collective debate (Haas, 1992). Haas's concept is broader than semantic proximity, involving shared normative and causal beliefs, validity criteria, and a common policy enterprise; the present study therefore does not operationalise an epistemic community directly, but measures only the weaker condition of observed semantic proximity. Semantic proximity cannot show the existence, strength, or influence of an expert community (Cross, 2013).

The benchmark itself is not neutral. The SDG framework is a politically negotiated instrument with documented limitations: goals can create incentives to over-focus on target achievement (Fukuda-Parr, 2016), summit documents remain state-centric and avoid causal responsibility (Bexell and Jönsson, 2017), and Goal 8's 3% annual growth target is empirically incompatible with the resource-use and emissions goals (Hickel, 2019). Reported SDG engagement is frequently superficial, so reported alignment should not be conflated with substantive commitment (Heilinger, Kempt, and Nagel, 2024; Heras-Saizarbitoria, Urbieta, and Boiral, 2022). To the extent that research follows Mode 1 knowledge production (problems set within academic disciplines) rather than Mode 2 (problems arising in contexts of application) (Gibbons et al., 1994), divergence is not uniformly problematic: where research explores possibilities that policy discourse has not yet absorbed, distance may reflect productive exploration rather than failure of alignment.

2.4 Semantic Methods: Rationale and Cautions

The distributional hypothesis, that meaning can be described through patterns of linguistic co-occurrence (Harris, 1954), underlies the use of Sentence-BERT (Reimers and Gurevych, 2019) and related transformer encoders for large-scale semantic comparison. Applying them to compare two distinct discourse corpora, research abstracts and policy documents, is a novel usage rather than a directly benchmarked one, and the interpretation of distance remains descriptive rather than causal. Gjorgjevikj et al. (2025) benchmarked sentence encoders specifically for SDG association tasks, finding that general-purpose models from the sentence-transformers family perform among the strongest as baselines, and that domain-specific fine-tuning further improves predictive performance while reducing sensitivity to description length. Rodriguez, Spirling, and Stewart (2023)'s conText framework provides a computational-social-science precedent for comparing meaning across contexts, adopting a narrow, structuralist view of meaning as distributional co-occurrence and making no causal claims about human understanding. This study follows that same descriptive premise: semantic proximity is a distributional measurement, not evidence of substantive framing alignment. The implication is that embedding distance mixes topic, register, and contextual convention into a single number: distributional representations are not topic-only, since language models inherently encode varieties of language including register (Grieve et al., 2025), and stylistic variation is demonstrably present in distributional vector spaces (Niu and Carpuat, 2017). This motivates the decomposition into these components (Section 3.8). McCarthy et al. (2026) supply a closer empirical precedent, showing with a multi-target SciBERT classifier that research–policy coverage gaps are measurable with transformer encoders outside the AI–SDG setting; this study extends that logic from a single conservation domain to the full SDG landscape, and from coverage to within-SDG semantic framing.

An older corpus-linguistic tradition adds a further caution. Biber's multidimensional (MD) analysis identifies six co-varying dimensions of linguistic variation across 23 spoken and written genres, derived from factor analysis of 67 linguistic features (Biber, 1988). Each dimension captures a continuum of co-occurring linguistic features rather than a binary speech/writing split (Biber, 1988; Biber and Conrad, 2009). Register is therefore a multi-layered property: two texts can differ on some dimensions and converge on others. The INLP procedure used in this study removes a *linear* subspace from the embeddings; the register component it isolates is best understood as capturing the dominant linear component of Biber's continuous register space, not as a claim that register is binary or that a single projection exhausts it. For the present study this matters because research abstracts and policy documents may differ not only in substantive topic, but also in abstraction, modality, institutional stance, and rhetorical function; embedding distance is therefore treated as observed textual-semantic proximity, not a clean measure of topic alone.

2.5 Research Gap

The literature reviewed above has not produced: (i) a systematic comparison of how academic AI-for-sustainability research and institutional SDG policy discourse differ in SDG coverage and within-SDG semantic framing; prior work measures research coverage, citation patterns, or policy-document uptake, but does not compare the two corpora directly on these dimensions; or (ii) an examination of whether coverage gaps and semantic gaps are positively related, weakly related, or inversely related. This framework is deliberately diagnostic: it maps where observed divergence is visible, without classifying whether any given gap reflects productive or problematic disconnection; the latter would require qualitative follow-up beyond the scope of embedding analysis.

These two contributions are motivated by the theoretical frameworks reviewed in Section 2.3. The two-communities theory holds that researchers and policymakers occupy separate worlds with different values, reward systems, and languages (Caplan, 1979). If these communities also develop different topical priorities, as the empirical record of research-policy divergence suggests (Strauss et al., 2025; McCarthy et al., 2026), then coverage divergence and semantic divergence should co-occur: SDGs where one community dominates attention should also show divergent framing, because the community shaping the discussion brings its own communicative norms. This motivates H1a. The epistemic-communities framework further distinguishes the *direction* of dominance: where research prioritises a given SDG, the framing should reflect academic norms of knowledge production; where policy prioritises it, the framing should reflect institutional norms of coordination (Haas, 1992). H1b–H1d decompose the coverage-semantic relationship into these directional components.

2.6 Research Questions and Hypotheses

The research question is operationalised by four sub-hypotheses, each correlating a distinct coverage predictor with the within-SDG semantic gap across the 17 SDGs:

- **H1a** (motivating hypothesis): the coverage gap is positively associated with the semantic gap;
- **H1b**: research–policy dominance $\leftrightarrow$ semantic gap (two-sided);
- **H1c**: research coverage $\leftrightarrow$ semantic gap (two-sided);
- **H1d**: policy coverage $\leftrightarrow$ semantic gap (two-sided).

H1b–H1d are signed tests of directional components: with dominance defined as research share minus policy share, the sign of each association locates which community's framing norms drive the linkage, per the epistemic-communities direction-of-dominance argument. SDG 4 is treated as artefact-affected due to ML-learning vocabulary overlap with education corpora, and is examined separately (Appendix B.1). The operationalization of these tests, Pearson r and Spearman ρ between each predictor and semantic-gap scores including the $n = 17$ scope conditions, is described in Section 3.9.

3 Methodology

3.1 Research Corpus

This study uses publicly available text data and requires no ethical approval.

The research corpus consists of 2,536,771 research abstracts retrieved from the OpenAlex API (Priem, Piwowar, and Orr, 2022), covering publications between 2018 and 2025. Retrieval combined OpenAlex's native SDG work filter with four high-precision AI search terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`), giving 68 queries (17 SDGs $\times$ 4 terms). The term set draws on established practice for delimiting AI corpora (Hellwig, Huggett, and Siebert, 2019; OECD, 2024), capturing a focused subset rather than the full universe of AI-related terms. Because this study measures semantic alignment between research and policy rather than counting publications, retrieval favours precision: each search term must clearly identify AI-related work, accepting that some relevant papers may be missed, to keep non-AI papers out of the embedding space. Complete query parameters are documented in Appendix A.

A direct test re-ran retrieval using field-of-study membership for the same two AI fields (OECD, 2024) and re-scored the resulting 100,000-paper corpus through the identical pipeline. Coverage and semantic-gap rankings are stable under this alternative strategy (Section 4.4; Appendix A).

An important asymmetry follows: the research corpus is a tight intersection of AI and SDG content (AI $\cap$ SDG), while the policy corpus (Section 3.2) is a broader union covering institutional SDG discourse and AI governance. Following Armitage, Lorenz, and Mikki (2020), the retrieval universe is a method-dependent corpus boundary. After deduplication and removal of records lacking usable abstracts, 2,536,771 abstracts were retained and tokenised into 3,105,144 segments (1.224 segments per abstract on average; 17.18% exceed the 374-token encoder window and split into multiple segments). A small number of records fail the 20-word segment minimum and are excluded: 2,543,698 records survive preprocessing, of which 6,927 are dropped at segmentation because their combined text is too short (mostly very short abstracts). The dropped share is below 0.3%. The corpus is predominantly English but no hard language filter is applied: roughly 0.5% of abstracts are predominantly non-English and are retained. The semantic-gap ranking reproduces under subsampling to 10,000 papers (Appendix E), two orders of magnitude larger than the non-English share. No balancing by SDG or citation count is applied.

3.2 Policy Corpus

The policy corpus contains 40,597 text segments from three collections, representing 6,367 unique source documents. It should be read as mixed *international institutional SDG policy discourse* rather than the full universe of AI sustainability policy:

1. **Curated AI and SDG policy documents** (13,123 segments, 94 documents): UN, OECD, EU, UK, and AU AI governance frameworks; IPCC assessment reports; SDSN sustainable development reports; and national AI strategies.
2. **SDGi Voluntary National Review and Voluntary Local Review corpus** (SDGi, 2024) (21,346 segments from 4,225 documents at segmentation; 20,267 single-label segments in the reference pool): National and local self-assessments submitted to the UN High-Level Political Forum.
3. **UN General Debate Corpus** (Jankin, Baturo, and Dasandi, 2024) (6,128 segments, 2,048 documents): SDG-relevant passages from General Debate speeches, sessions 70–80 (2015–2024); paragraph-level keyword filtering precedes segmentation, so only SDG-relevant passages enter the pipeline, which is why its 6,128 segments are far fewer than the full corpus of speeches.

The curated sub-corpus (94 documents) is the more relevant reference for AI-governance-specific interpretation; Appendix H.6 reports a source-family sensitivity check.

3.3 Token-Aware Segmentation

All source documents are segmented at sentence boundaries using the embedding model's own tokenizer to enforce a hard token budget, guaranteeing that every segment fits within the encoder's context window. The procedure applies identically to research abstracts, policy documents, and reference corpora: NLTK sentence-boundary detection is followed by greedy accumulation of sentences up to a token budget of `max_seq_length` -10 (for the canonical MPNet encoder, $384 - 10 = 374$ tokens). Segments falling below 20 words are discarded; a single sentence exceeding the token budget is kept whole because mid-sentence splitting is not permissible.

For the research–policy comparison this design has two consequences. First, research abstracts (mean $\sim$200 words) mostly fit within the 374-token budget, but 17.18% exceed it and split into multiple segments; the research-side text unit is therefore the abstract, whose segments are aggregated to a single document vector before assignment, not the individual segment. Second, policy documents of variable length are split into multiple segments of comparable size to research abstracts, making the two corpora directly comparable in embedding space. The per-document segment cap of $K = 50$ applied in the semantic-gap analysis (Section 3.7) is a separate, additional restraint on document-level dominance within a single SDG cluster; it does not affect the segmentation itself.

3.4 Embedding Model and Normalisation

All texts were encoded with `all-mpnet-base-v2` (Sentence-Transformers, 2021b; Reimers and Gurevych, 2019) from the sentence-transformers library, producing 768-dimensional dense vectors. Encoder choice is a principal robustness axis: gap rankings are re-computed under two alternative encoders, all-MiniLM-L6-v2 (Sentence-Transformers, 2021a) (384-dimensional) and SciBERT (Beltagy, Lo, and Cohan, 2019) (768-dimensional). Embeddings were L2-normalised to unit vectors so that cosine similarity equals their dot product.

No fine-tuning is applied: under the linear-probing paradigm (Alain and Bengio, 2018; Conneau et al., 2018), a linear classifier on frozen embeddings is the standard instrument for checking whether a representation already encodes the signal of interest, and the supervised reference classifier (Section 3.5) and the INLP register projection (Section 3.8) are both linear instruments on this frozen space. Freezing is also a measurement requirement: the two corpora must be compared in one fixed, shared space, and fine-tuning on SDG reference data, which are weighted toward policy vocabulary, would re-shape that space and confound the comparison. A frozen pipeline also carries no fine-tuning randomness, keeping the analysis deterministic and reproducible.

3.5 Supervised Reference Classifier

The canonical instrument is a supervised multinomial logistic regression (LR) classifier trained on pooled single-label reference data from five independent annotated corpora (counts below are single-label segments after preprocessing):

- **OSDG Community Dataset** (Pukelis et al., 2022) (30,532 texts, SDGs 1–16): crowdsourced excerpts validated by multiple volunteers.
- **SDG Classification Benchmark** (SDG Classification Expert Group, 2025) (281 expert-verified texts, SDGs 1–17): expert-annotated snippets with consensus resolution.
- **IISD SDG Knowledge Hub** (Wulff, Meier, and Mata, 2023) (6,122 texts, SDGs 1–17): journalism and policy commentary on 2030 Agenda implementation.
- **SDGi Corpus** (SDGi, 2024) (20,267 texts, SDGs 1–17): government Voluntary National and Local Reviews.
- **Aurora dataset** (Vanderfeesten, Spielberg, and Gunes, 2020) (4,971 expert-validated texts, SDGs 1–17): academic abstracts validated by domain researchers.

OSDG alone contributes about 49% of labelled texts and draws on UN-adjacent documents, so the reference data remain weighted toward policy vocabulary. OSDG and the Benchmark bypass segmentation; the Knowledge Hub, SDGi, and Aurora are segmented before pooling.

The pooled reference data (n=62,173 after single-label filtering) were split into training and held-out test sets (n=52,835 train, n=9,338 test) by stratified document-grouped splitting, keeping all texts from the same source document in one fold to enforce strict separation between training and evaluation. The champion LR ($C = 3.0$, L2 penalty, `class_weight=None`, solver `lbfgs`) was selected by 5-fold GroupKFold cross-validation (Hastie, Tibshirani, and Friedman, 2009) on the training pool and reaches macro-$F_1 = 0.816$ on the held-out test set (Section 4.1). The grid-search protocol, hyperparameter rationale, and the MLP robustness comparison are reported in Appendix F. The validation task does not evaluate a rejection option for non-SDG texts; this is acceptable because the research corpus is first restricted to an OpenAlex AI-and-SDG universe, so the classifier answers a relative question among the 17 SDGs.

Each text is assigned to its highest-probability SDG (argmax), traceable to 768 signed coefficients per class with no hidden layers, no dropout, and no random-seed sensitivity. Macro-F_1, the unweighted mean of per-SDG scores, is the primary validation metric (Section 4.1).

Because the classifier is trained on pooled reference data that are themselves policy-adjacent, the scoring instrument behaves differently on the two corpora. Policy segments receive systematically higher LR predicted probabilities than research abstracts (mean

0.749 vs 0.627, a 0.122 gap). The gap has two causes the data cannot separate: a genre artefact (the reference data skew toward policy vocabulary) or a substantive difference (policy texts address SDG implementation more directly). Either way the asymmetry does not threaten the findings: hard argmax assignment uses only within-corpus ranking, not absolute probability levels, so the directional gap does not reorder goals. It is not an SDGi circularity effect: the training set and scored policy corpus are disjoint by construction.

3.6 Coverage Gap Analysis

For each corpus item, the predicted SDG is the LR argmax (hard assignment, Section 3.5); the coverage profile is the proportion of analysis units assigned to each SDG (an abstract for research, a source document for policy).

The policy corpus is not a flat set of documents: its 40,597 segments are grouped into 6,367 source documents of very different lengths (Section 3.2). A segment-weighted profile would let a handful of long reports dominate the count, so the analysis instead computes a *document-weighted* profile in which every source document counts equally regardless of length, following the standard centroid formulation in which each document contributes one length-normalised unit vector (Manning, Raghavan, and Schütze, 2008). Concretely: segment vectors are averaged per document, hard-assigned to the top SDG, and the coverage profile counts documents (not segments) per SDG. The research corpus uses the same logic at the paper level: each of the 2,536,771 abstracts is represented by the mean of its segments, hard-assigned as a single document unit, so the research-side text unit is the abstract and the two corpora are weighted identically.

The coverage gap for SDG j is defined as:

$$\text{CoverageGap}_j = \left|\text{ResearchCoverage}_j - \text{PolicyCoverage}_j\right|$$

The signed version of the same quantity is:

$$\text{SignedCoverageGap}_j = \text{ResearchCoverage}_j - \text{PolicyCoverage}_j$$

A positive value means research dominates SDG j; a negative value means policy dominates.

3.7 Semantic Gap Analysis

The semantic gap measures within-SDG divergence: given that both corpora have texts assigned to SDG j, how differently do they discuss it? For each SDG j, the analysis collects all assigned research papers and all assigned policy segments (capped at 50 per source document to bound single-document influence; SDSN reports alone contribute up to 3,179 segments otherwise). The two corpora are aggregated at different text units here, by design—unlike the coverage profile (Section 3.6), which weights both corpora at the document level. Research abstracts are single-topic by genre convention, so each of the 2,536,771 abstracts contributes exactly one unit vector, the L2-renormalised mean of its segment embeddings, and a 33-segment abstract and a single-segment abstract carry equal weight: the research centroid $\mathbf{c}_j^{\text{res}}$ is the L2-normalised mean of these paper-level unit vectors assigned to SDG j. Policy sources, by contrast, are long, deliberately excerpted institutional texts (UNGDC speeches, SDGi VNRs) whose passages are topically distinct and may span several SDGs; weighting each source document equally would collapse that within-document variation into a single point. The policy centroid $\mathbf{c}_j^{\text{pol}}$ is therefore the L2-normalised mean of the (capped) segment embeddings assigned to SDG j, sampling the topical content of each passage rather than treating each report as one undifferentiated unit. Underlying both choices is the same rule: the coverage profile counts documents, so both corpora are aggregated to the document level; the semantic gap averages text content, so each corpus is aggregated at its natural authorship unit—the abstract for research and the passage for policy. The semantic gap for each SDG is then defined as:

$$\text{SemanticGap}_j = 1 - (\mathbf{c}_j^{\text{res}} \cdot \mathbf{c}_j^{\text{pol}})$$

A gap of 0 indicates perfectly overlapping semantic directions; a gap approaching 1 indicates orthogonal framing. Cosine similarity between centroids is the standard method for document-level semantic comparison in embedding space (Brück and Pouly, 2019; Babićc et al., 2020). Centroid cosine distance collapses each embedding cloud to its mean direction. Appendix D shows the adjusted-gap ranking correlates moderately with most full-distribution alternatives ($\rho = 0.44$–0.69), while Gaussian-2-Wasserstein, squared MMD, Sinkhorn, C2ST, and Grassmann show weak or negative agreement ($\rho \leq 0.26$), confirming the centroid estimate captures the dominant signal while distribution shape carries a related but distinct component.

The *per-document segment cap* of $K = 50$ (seeded, reproducible) bounds a single document's contribution to at most 50 segments, mitigating—but not fully equalising—the influence of very long reports on the policy centroid; sensitivity at $K = 20$ and no cap is reported in Table 17.

3.8 Register Adjustment via Iterative Nullspace Projection (INLP)

The semantic gap is not a clean measure of substantive disagreement. Following Biber and Conrad's distinction between register, genre, and style (Biber and Conrad, 2009), this study treats the research–policy contrast as a composite of register and topic: the two corpora differ in situation of use, audience, and communicative purpose (register), and also in substantive content (topic). Embedding distance therefore mixes genuine within-SDG substantive divergence with register divergence in abstraction, modality, and institutional stance (Biber, 1988). To separate these components, the semantic gap is decomposed into an adjusted semantic gap and a register component using Iterative Nullspace Projection (Ravfogel et al., 2020, INLP). After register removal, the adjusted gap is the primary estimate of within-SDG topic divergence, while the residual register component captures the removed register signal (Table 2).

Objective. Following Ravfogel et al. (2020), the goal is to learn a linear projector that removes corpus-identity information from embeddings while preserving distances as much as possible. After K iterations the composite projector P removes all linearly decodable corpus signal; the orthogonal nullspace projection is the least harmful linear operation for this purpose.

SDG-stratified procedure. Vanilla INLP removes a binary attribute (e.g. gender). The present adaptation must remove *register*, a corpus-level property that is confounded with SDG topic in the raw embeddings. The key modification is SDG stratification: at each iteration, equal numbers of research and policy embeddings are drawn from each of the 17 SDGs, preventing the classifier from learning topic membership as a proxy for register. The procedure proceeds as follows at each iteration $k = 1, \ldots, K$:

1. **Stratified sample.** Draw n_{target} texts per SDG per corpus (global-minimum rule: $n_{\text{target}} = \min_j \min(|X_j^{\text{res}}|, |X_j^{\text{pol}}|)$; under the frozen snapshot, $n_{\text{target}} = 1123$ for MPNet). Form the 34-class stratification key $z_{\text{strat}} = 2 \times \text{sdg\_labels} + y$, ensuring balanced corpus × topic combinations.
2. **Project existing directions.** Subtract from each sample embedding the components along all previously learned directions: $\tilde{\mathbf{x}} = \mathbf{x} - \sum_{j<k} (\mathbf{g}_j \cdot \mathbf{x})\, \mathbf{g}_j$.
3. **Train classifier.** Fit L2-regularised logistic regression to predict corpus identity y from the projected embeddings $\tilde{\mathbf{x}}$; the hyperparameters are fixed defaults, not validated, because the objective is to recover a decodable direction, not to maximise classification accuracy. Split 15% held-out, stratified by z_{strat}.
4. **Extract direction.** Normalise the weight vector: $\mathbf{g}_k = \mathbf{w}_k / \|\mathbf{w}_k\|$. Orthogonalise against all prior directions: $\mathbf{g}_k \leftarrow \mathbf{g}_k - \sum_{j<k} (\mathbf{g}_k \cdot \mathbf{g}_j)\, \mathbf{g}_j$. If $\|\mathbf{g}_k\| < 10^{-8}$ (direction collapse), stop.
5. **Stopping criterion.** Stop if held-out accuracy ≤ 0.5 (the majority-class baseline for balanced binary classification). Otherwise append $\mathbf{g}_k$ and continue.

All three encoder tracks converge within 40–79 iterations (Appendix G.1). The final adjusted embedding is $\mathbf{x}' = \mathbf{x} - \sum_{k=1}^{K} (\mathbf{g}_k \cdot \mathbf{x})\, \mathbf{g}_k$, and the adjusted gap uses centroids computed on these projected embeddings.

Identification argument. The stratification design is the critical structural precondition: because every iteration trains on equal numbers of research and policy embeddings from each SDG, no linear direction can simultaneously encode SDG topic and corpus identity. What remains linearly decodable is the corpus-level register difference: modality, audience conventions, institutional stance, and structural patterns. The removed subspace is interpreted as register in the sense of Biber and Conrad (2009): a variety associated with a particular situation of use, including communicative purpose, audience, and production circumstances. Because stratification balances SDG topic but not other variety axes, the removed component may combine register with period effects and author-background signals, so the register reading is a supported interpretation rather than an exhaustive characterisation (Grieve et al., 2025). This interpretation is evaluated against independent linguistic markers in Appendix G.2; the validation finds substantial reduction of the corpus-level signal but not its complete elimination.

With the projector learned, the semantic-gap analysis of Section 3.7 is recomputed twice: once on the adjusted embeddings and once on the removed component ($\mathbf{x} - \mathbf{x}'$), yielding respectively the within-SDG *adjusted gap* and the *register component*.

3.9 Coverage–Semantic Interaction

Their operationalization is as follows. Pearson r and Spearman rank correlation ρ (Spearman, 1904) are computed between each coverage predictor (coverage gap, research–policy dominance, research coverage, policy coverage) and the within-SDG semantic-gap scores across the 17 SDGs; SDG 4 is treated as artefact-affected and examined separately (Appendix B.1).

The 17 SDGs are the complete framework, not a sample; the observed associations are properties of the framework, and the only stochastic element is the measurement process (classifier assignment, corpus construction, embeddings). All reported p-values are two-sided Monte Carlo permutation p-values (100,000 resamples, fixed seed): they condition on the measured values and test the null of no association between them, so no sampling-based power reasoning is involved (Berk, Western, and Weiss, 1995; Abadie et al., 2020).

The operative small-n concern is robustness rather than power: with only 17 framework units, individual goals can be influential, so the associations are read together with the leave-one-out check (Section 4.3) and the cross-configuration checks (Section 4.4).

The raw, adjusted, and register-component correlations are a single decomposition (register = raw − adjusted), mechanically linked; the substantive claim concerns their joint cancellation pattern, not three separate significance decisions. All three are reported unconditionally for every configuration, so the family is fully disclosed. Because the decomposition is pre-specified (Section 3.8) and the statistics are jointly informative, no familywise correction is applied.

Finer-grained units (the framework defines 169 targets) would raise the observation count, but no labelled target-level reference data exist. A supplementary pooled-regression analysis combines all 24 configurations into a single panel (Appendix I).

Figure 2 summarises the end-to-end methodology.

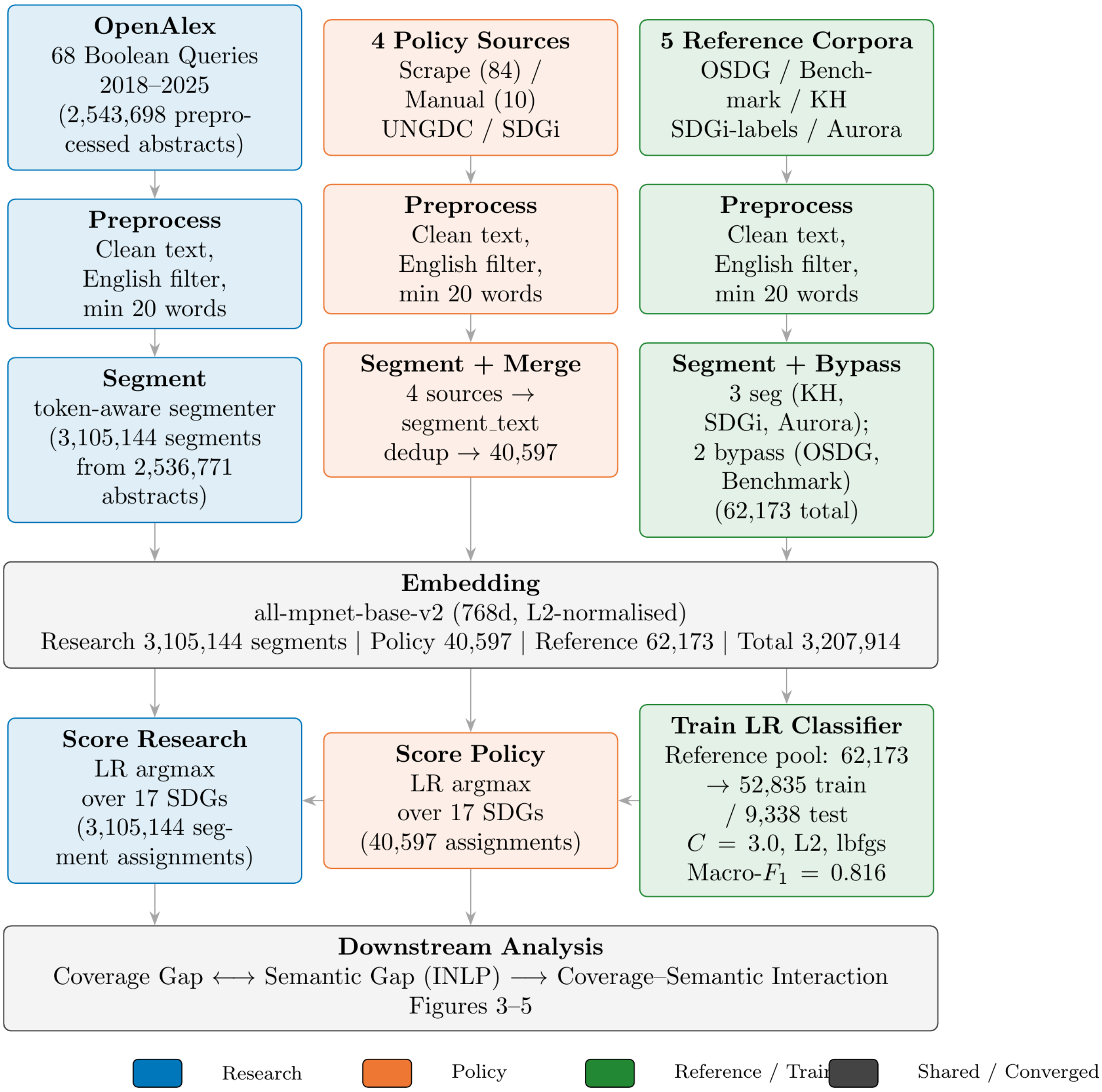

Figure 2. End-to-end methodology pipeline in three parallel data tracks.

4 Results

This section reports validation, coverage gaps, semantic gaps, H1 tests, and robustness without interpretation.

4.1 Supervised Reference Classifier and Coverage Gap

The LR classifier was evaluated on the held-out test set (n=9,338, document-grouped to prevent source leakage). It reaches macro-F_1 = 0.816 and micro-F1 = 0.8306, far above the 1/17 random baseline (0.059). Per-class F1 values from the test set are reported in Table 1.

Table 1. LR test-set F1 scores (n=9,338).

SDG	LR test F1
1 (No Poverty)	0.737
2 (Zero Hunger)	0.830
3 (Good Health)	0.903
4 (Education)	0.889
5 (Gender Equality)	0.881
6 (Clean Water)	0.879
7 (Clean Energy)	0.861
8 (Decent Work)	0.716
9 (Innovation)	0.770
10 (Reduced Ineq.)	0.610
11 (Sust. Cities)	0.762
12 (Consumption)	0.783
13 (Climate)	0.831
14 (Life Below Water)	0.892
15 (Life on Land)	0.865
16 (Peace & Justice)	0.879
17 (Partnerships)	0.788
Macro-F1 (SDGs 1–17)	**0.816**

Notes: LR test F1 on the held-out pooled test set (all reference sources). The table shows the canonical 17-way classifier used throughout the main analysis.

Per-class F1 ranges from 0.610 (SDG 10) to 0.903 (SDG 3); most goals score between 0.76 and 0.89. The weakest classes (SDGs 10 and 8) are consistent with their semantic breadth (Kashnitsky et al., 2024) (Appendix B.2, Figure 7).

The MLP robustness check achieves comparable performance (MLP: 0.826; LR: 0.816), confirming architecture choice is not the binding constraint (Appendix F).

Figure 3 presents the document-weighted policy coverage profile alongside the research coverage profile.

Policy corpus. Three SDGs dominate the document-weighted profile of the full institutional policy corpus: SDG 16 (Peace, Justice and Strong Institutions, 17.3%), SDG 13 (Climate Action, 11.3%), and SDG 17 (Partnerships for the Goals, 11.7%), accounting for 40.3% of policy documents. This ordering is consistent with the literature review's account of SDG 17 as the cross-cutting means-of-implementation goal (Section 2) (Le Blanc, 2015; D. J. Griggs et al., 2017). The SDG 16 lead is notable: under the supervised classifier, governance and justice discourse emerges as the most prevalent policy theme, ahead of both climate and partnerships. Appendix H.6 shows this full-corpus profile is source-family

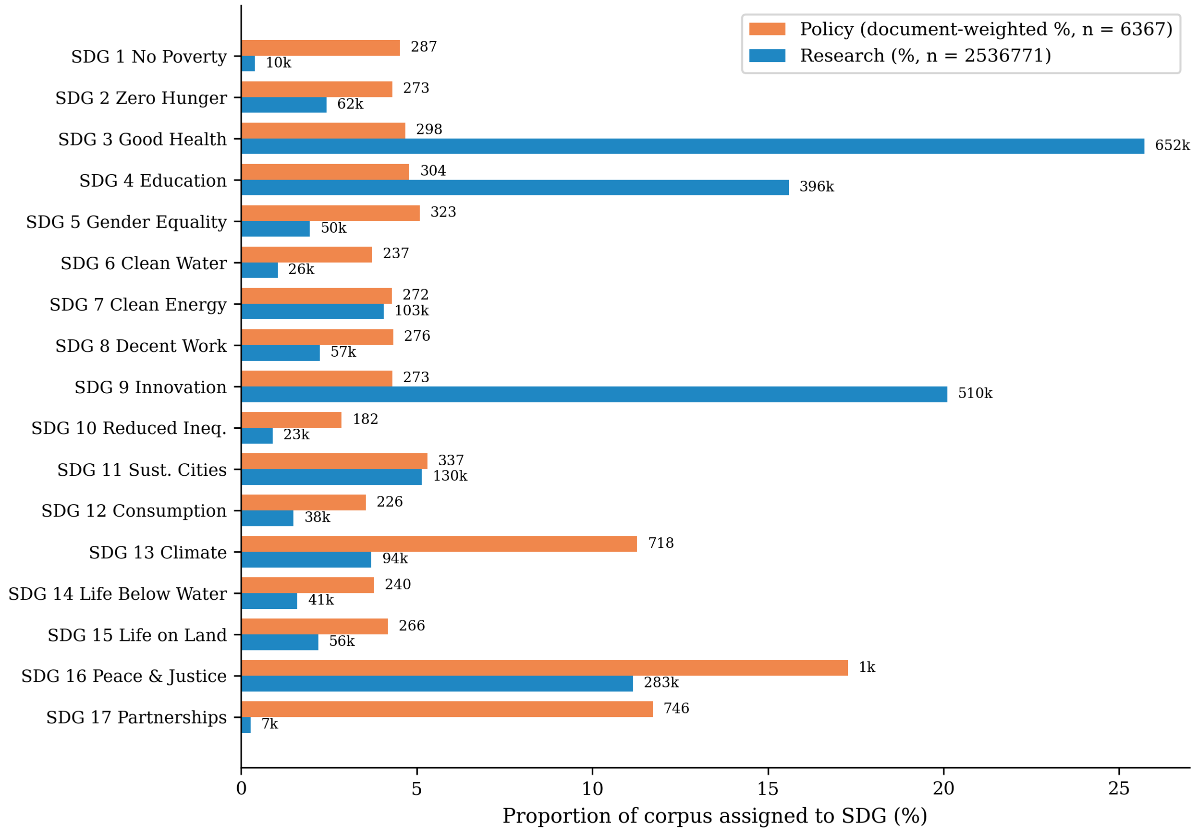

Figure 3. Coverage profiles by SDG. SDG 4 is flagged as artefact-affected due to ML-learning vocabulary overlap with education corpora (Appendix B.1).

sensitive: the curated AI/SDG sub-corpus is much more innovation-heavy, the UNGDC family is more climate-, partnerships-, and governance-oriented, and the SDGi family is comparatively flat, with SDG 11-style institutional review language at its top.

Research corpus. The research profile is broader, but it still concentrates strongly on a smaller set of SDGs. The largest research shares are SDG 3 (Good Health, 25.7%), SDG 9 (Innovation, 20.1%), and SDG 4 (Quality Education, 15.6%), followed by SDG 16 and SDG 11 at 11.2% and 5.1% respectively. SDG 7 (Affordable and Clean Energy), co-dominant in several prior inventories, ranks sixth here (4.1%) rather than in the leading tier. The SDG 3 lead corroborates the literature review, where SDG 3 is the one goal every major AI and sustainability bibliometric study finds dominant Bautista-Puig et al. (2021), Singh et al. (2024), Cowls et al. (2021), and Nedungadi et al. (2024), and Gohr et al. likewise show health among the largest SDG-focus shares in their reviewed corpus (Gohr et al., 2025). The SDG 4 share is the one point of disagreement: prior inventories do not rank SDG 4 among the leaders, so its third-place finish here is the same result flagged as likely inflated by machine-learning vocabulary (Appendix B.1). Even so, the research corpus clearly does not mirror the policy-side dominance of SDGs 16, 13, and 17; the research-side near-absence of SDG 17 (0.3%) and SDG 1 (0.4%) matches Gohr et al.'s finding that SDG 1, SDG 5, and SDG 17 are severely underrepresented in the AI–SDG literature (Gohr et al., 2025).

Largest coverage mismatches. The five largest absolute coverage gaps are SDG 3 (0.210), SDG 9 (0.158), SDG 17 (0.114), SDG 4[1] (0.108), and SDG 13 (0.076). The largest single-goal imbalance is SDG 3 (research 25.7% vs. policy 4.7%): research concentrates on technically tractable health applications while policy underweights the goal. The most extreme policy-over-research imbalances are SDG 17 (policy 11.7% vs. research 0.3%) and SDG 13 (11.3% vs. 3.7%): policy frames partnerships and climate as cross-cutting coordination challenges, while research engages them far less. SDG 16 is policy's top theme (17.3%) but research does engage it (11.2%, third-highest research share), so it is not a "barely-touches" case. The SDG 17 mismatch is corroborated independently: Gohr et al. found no SDG 17 articles at all among the 792 AI–SDG papers they reviewed (Gohr et al., 2025).

4.2 Semantic Gap after Register Removal

Figure 4 shows the combined embedding space and the geometric effect of INLP register removal: the raw space (left) separates research and policy clouds, while the adjusted space (right) merges them, confirming that the separation is driven by register rather than topic. The policy cloud hugs the 17 reference centroids more tightly than the research cloud. The projection is descriptive only (9.2% and 3.7% of variance); cosine distances remain in the original 768-dimensional space. The effect is encoder-dependent, since SciBERT already fuses the clouds without INLP, so this figure is MPNet-specific.

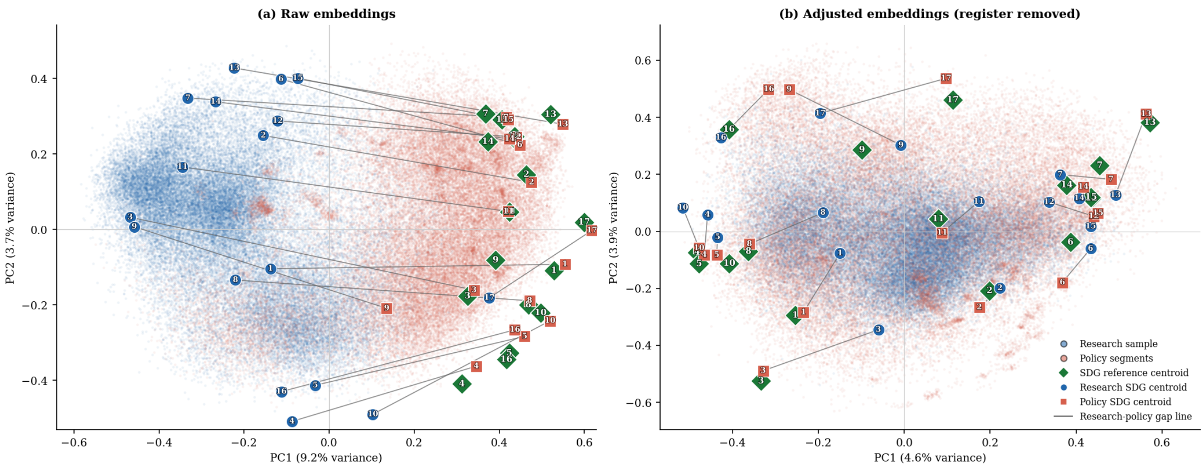

Figure 4. PCA before/after INLP register removal (MPNet).

Figure 5 shows the within-SDG semantic gap for all 17 SDGs, sorted by the adjusted (topic) gap after register removal. The adjusted gap ranges from 0.326 (SDG 17, widest) to 0.084 (SDG 15, narrowest), a span of 0.241 cosine units. The raw gap (naive baseline)

[1]SDG 4's share is partly inflated by machine-learning vocabulary overlap (Appendix B.1).

ranges from 0.494 (SDG 3, widest) to 0.177 (SDG 17, narrowest), a span of 0.317. The adjusted ranking is stable when the segment cap is 20 per document or removed (Table 17), so the result does not depend on the sampling parameter. A second sampling concern is corpus asymmetry: research (2,536,771 abstracts) outweighs policy (6,367 documents) by roughly 398:1 in document units. This study therefore re-derived the ranking on random research draws comparable in size to the policy corpus: across 100 independent 50,000-paper draws it reproduces the full-corpus ordering at Spearman $\rho = 0.986$ (SD 0.009), so the research-side over-representation does not drive which SDGs show large versus small gaps (Appendix E).

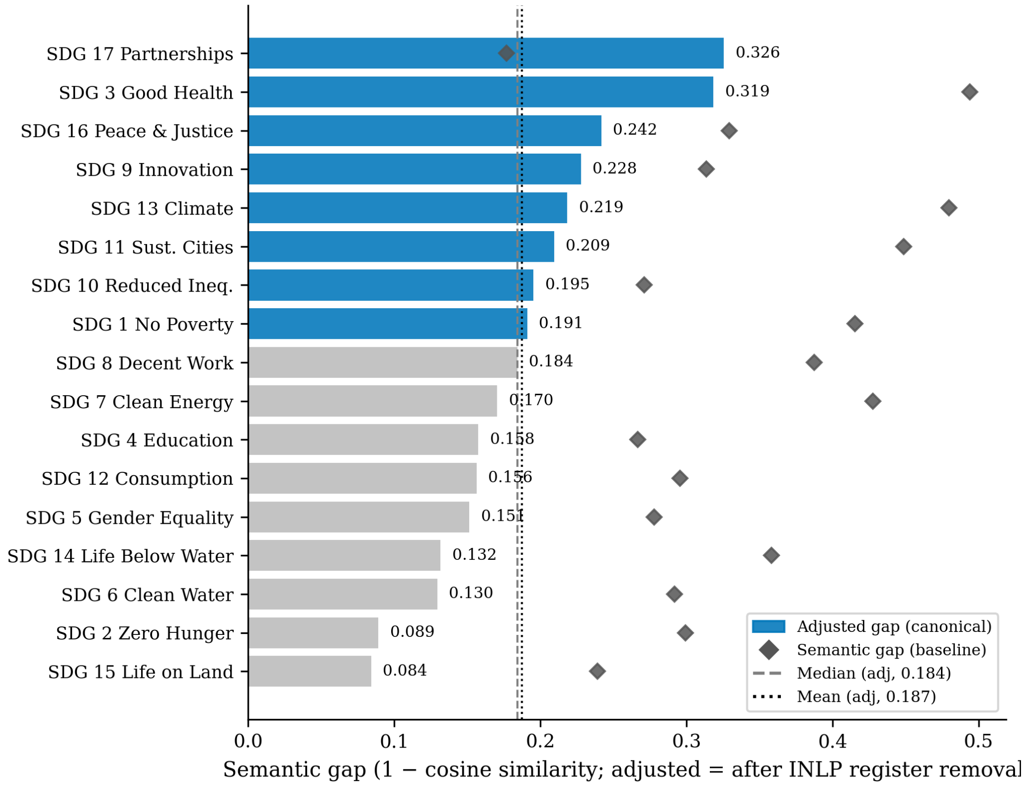

Figure 5. Within-SDG semantic gap by SDG, sorted by the adjusted (topic) gap after register removal.

Largest semantic gaps (adjusted, primary). SDGs 17, 3, 16, 9, and 13 show the widest adjusted gaps. Under the raw gap, the ordering shifts: SDG 3 (0.494) leads. The register decomposition confirms that register uniformly inflates the raw gap, except for SDG 17 where register *increases* apparent similarity.

Smallest semantic gaps (adjusted, primary). SDG 15 (Life on Land, 0.084), SDG 2 (Zero Hunger, 0.089), SDG 6 (Clean Water, 0.130), SDG 14 (Life Below Water, 0.132), and SDG 5 (Gender Equality, 0.151) show the narrowest adjusted gaps. A narrow gap does not mean balanced attention: SDG 9 is still research-dominant, and SDG 4 carries the learning-vocabulary concern (Appendix B.1). Appendix C lists the most distinctive

Table 2. Register-topic decomposition of the semantic gap.

SDG	Description	Semantic Gap	Adj. Gap	Reg. Comp.	Cov. Gap
1	No Poverty	0.415	0.191	+0.224	0.0413
2	Zero Hunger	0.299	0.089	+0.210	0.0187
3	Good Health	0.494	0.319	+0.176	0.2104
4	Education	0.267	0.158	+0.109	0.1081
5	Gender Equality	0.278	0.151	+0.126	0.0314
6	Clean Water	0.292	0.130	+0.162	0.0269
7	Clean Energy	0.428	0.170	+0.257	0.0023
8	Decent Work	0.387	0.184	+0.203	0.0210
9	Innovation	0.314	0.228	+0.086	0.1580
10	Reduced Ineq.	0.271	0.195	+0.076	0.0196
11	Sust. Cities	0.449	0.209	+0.239	0.0017
12	Consumption	0.296	0.156	+0.139	0.0206
13	Climate	0.480	0.219	+0.261	0.0757
14	Life Below Water	0.358	0.132	+0.227	0.0219
15	Life on Land	0.239	0.084	+0.155	0.0198
16	Peace & Justice	0.329	0.242	+0.087	0.0611
17	Partnerships	0.177	0.326	-0.149	0.1145
Mean	—	0.340	0.187	+0.152	0.0561

Notes: Raw gap is the reference semantic-gap estimate; Adjusted gap is after INLP register removal (primary estimate; evaluated in Appendix G.2); Register component = raw − adjusted. Coverage correlations are Spearman ρ across 17 SDGs.

research- and policy-side terms for high- and low-gap cases.

4.3 Coverage–Semantic Interaction: Results

Table 3 reports H1a–H1d rank correlations between coverage predictors and the within-SDG semantic gap across all encoder–classifier configs. Columns show Spearman ρ against the raw gap, adjusted gap, and register component.

Predictors are research coverage, policy coverage, absolute coverage gap, and signed dominance. MLP configs reuse their encoder's INLP matrix; Concept rows use the concept-retrieved research corpus (Appendix A).

H1a is null on the raw gap ($\rho = -0.012$, $p = 0.959$; Table 3, MPNet LR). The raw gap masks a register–topic decomposition: the register component ($\rho = -0.390$) cancels the topic signal ($\rho = 0.544$). After register removal, the adjusted gap is positively associated with the coverage gap (positive in 8/9 configs). The directional decomposition reveals an asymmetric pattern: H1d (policy coverage) is positive in 9/9 configurations, significant at $p < 0.05$ for MPNet LR ($\rho = +0.589$), SciBERT LR ($\rho = +0.605$), and Concept LR ($\rho = +0.510$), while H1b (signed dominance) and H1c (research coverage) show no consistent adjusted-direction signal. The linkage between coverage and framing operates through the policy side: where policy heavily covers an SDG, topic divergence is wider.

Table 3. H1a–H1d coverage-predictor vs semantic-gap rank correlations (Spearman ρ) across encoder–classifier configs.

	Semantic Gap	Adj. gap	Reg. Comp.
H1a. Coverage gap ↔ Semantic gap			
MPNet LR	-0.012	+0.544*	-0.390
MPNet MLP	-0.135	+0.336	-0.370
MPNet ZS	-0.299	+0.203	-0.449†
MiniLM LR	-0.243	+0.397	-0.463†
MiniLM MLP	-0.042	+0.505*	-0.272
SciBERT LR	-0.130	+0.221	-0.267
SciBERT MLP	-0.333	-0.034	-0.120
Concept LR	+0.108	+0.434†	-0.194
Concept MLP	-0.017	+0.181	-0.140
H1b. Policy–research dominance ↔ Semantic gap			
MPNet LR	+0.159	-0.059	+0.029
MPNet MLP	+0.020	-0.118	-0.034
MPNet ZS	-0.005	-0.336	+0.360
MiniLM LR	+0.404	-0.020	+0.429†
MiniLM MLP	+0.319	-0.066	+0.348
SciBERT LR	+0.549*	-0.118	+0.547*
SciBERT MLP	+0.336	-0.429†	+0.529*
Concept LR	+0.051	-0.029	-0.260
Concept MLP	-0.174	+0.093	-0.199
H1c. Research coverage ↔ Semantic gap			
MPNet LR	+0.449†	+0.267	+0.145
MPNet MLP	+0.262	+0.098	+0.088
MPNet ZS	-0.100	-0.328	+0.314
MiniLM LR	+0.444†	+0.262	+0.331
MiniLM MLP	+0.373	+0.275	+0.279
SciBERT LR	+0.605*	+0.466†	+0.020
SciBERT MLP	-0.005	-0.196	+0.225
Concept LR	+0.221	+0.017	-0.076
Concept MLP	-0.098	-0.010	-0.127
H1d. Policy coverage ↔ Semantic gap			
MPNet LR	+0.212	+0.589*	-0.009
MPNet MLP	+0.175	+0.428†	-0.007
MPNet ZS	-0.094	+0.240	-0.321
MiniLM LR	-0.015	+0.449†	-0.159
MiniLM MLP	+0.083	+0.456†	-0.093
SciBERT LR	-0.100	+0.605*	-0.686**
SciBERT MLP	-0.172	+0.395	-0.395
Concept LR	+0.369	+0.510*	+0.178
Concept MLP	+0.256	+0.258	+0.082

Notes: Raw gap = main semantic-gap estimate; Adj. gap = after INLP register removal (main reported estimate; evaluated in Appendix G.2); Register = raw − adjusted. Each correlation is between two rank vectors (ranks 1–17 across the 17 SDGs); Spearman ρ is the Pearson correlation of these ranks, testing monotonic association. $^{***}p < .001$, $^{**}p < .01$, $^{*}p < .05$, $^{\dagger}p < .10$.

The adjusted association is marginal: under a conservative familywise view of the three linked statistics, $p = 0.025$ would not survive a correction, so it is read as pattern-consistent evidence for the cancellation rather than a confirmatory test, with the weight carried by the cross-configuration and cross-encoder consistency above (Section 4.4).

A metric-sensitivity check repeats this grid with Pearson r on the raw magnitudes instead of Spearman ρ (Appendix H.5); the adjusted-gap association is unchanged in sign under the metric switch for the signal-bearing hypotheses (H1a, H1d); the few sign changes are confined to insignificant configs in the null H1b/H1c, confirming the rank-based reporting is not driving the result.

The SDG 4 lexical artefact (Appendix B.1) does not drive this result. A leave-one-out check with SDG 4 removed leaves the raw coverage-gap correlation null ($\rho = 0.088$, $p = 0.744$) and preserves the positive register-adjusted coverage-gap association ($\rho = 0.597$, $p = 0.016$), so the artefact props up neither the raw null nor the adjusted association.

The corpus asymmetry does not manufacture this null. Because the research corpus is the AI $\cap$ SDG intersection while the policy corpus pools AI-specific and broad institutional SDG discourse, this study re-ran the interaction test with the policy side restricted to the curated AI/SDG family, a symmetric AI/SDG-vs-AI/SDG comparison that matches the research corpus's scope. The cancellation pattern holds: research share remains uncorrelated with the gap ($\rho = -0.06$, $p = 0.833$), and the coverage-gap correlation does not reach significance either ($\rho = -0.22$, $p = 0.386$; Table 28, Appendix H.6).

Figure 6 shows adjusted semantic gap vs. coverage gap; open markers show the raw baseline. Large coverage gaps span the full range of adjusted semantic gaps; no single SDG drives the weak H1a association.

4.4 Robustness of the Gap Rankings

Register removal is validated against independent linguistic markers (Appendix G.2); the adjusted gap is the primary estimate.

Cross-sensitivity (policy source, segment cap, retrieval). Under MPNet–LR, the semantic-gap ranking is stable across segment cap and retrieval in the adjusted panel ($\rho \geq 0.65$; Table 17) and across policy source, segment cap, and retrieval in the raw panel ($\rho \geq 0.65$; Table 18). The same holds for coverage (Table 22).

Encoder sensitivity. Coverage-gap ranks are stable across encoders (MPNet–LR vs. SciBERT–LR $\rho = 0.93$; vs. MiniLM–LR $\rho = 0.89$; Table 21). Semantic-gap magnitude is encoder-dependent (SciBERT: 0.04 vs. MPNet: 0.34) because scientific pretraining fuses the genres (Beltagy, Lo, and Cohan, 2019), but the divergence conclusion holds (Table 19). Under the raw gap, SDG 3 leads for MPNet–LR and SciBERT–LR while SDG 13 leads

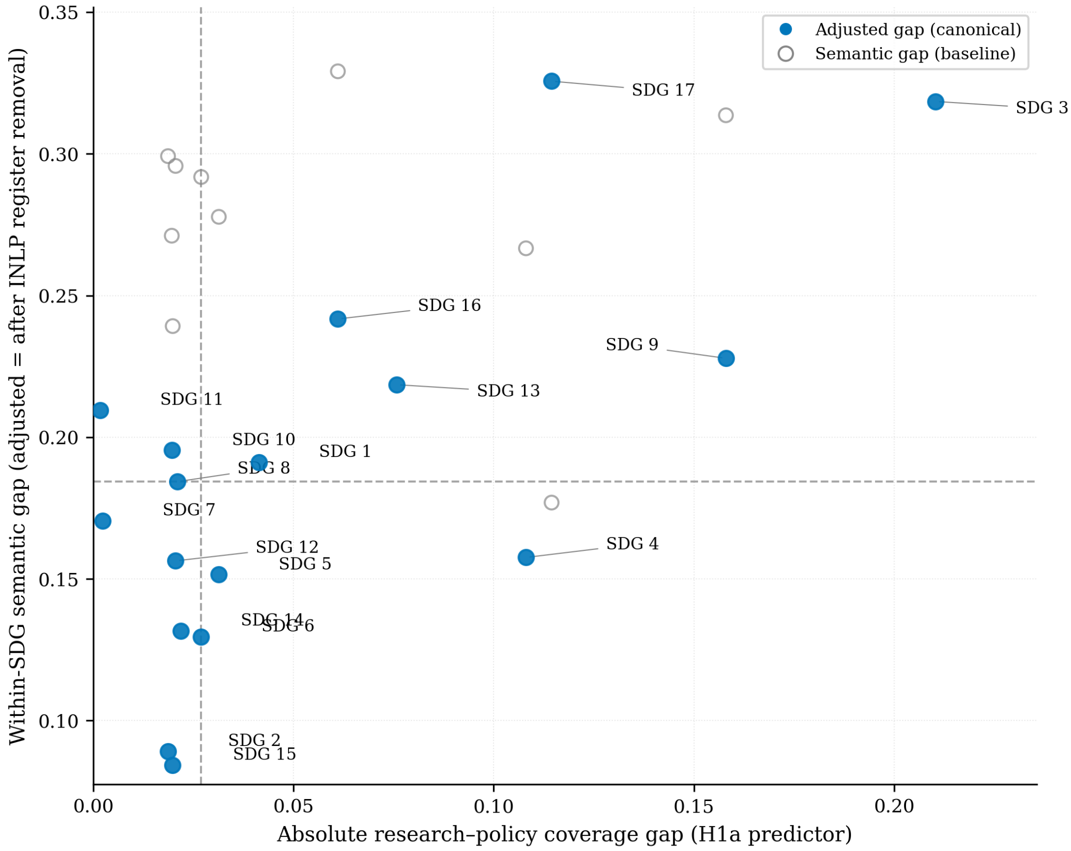

Figure 6. Coverage gap vs semantic gap, one point per SDG.

for MiniLM–LR; SDG 17 trails in all three (rank 16 or 17).

Classifier sensitivity. LR and MLP show varied agreement (MPNet LR vs. MLP $\rho = 0.85$; Table 19). SDG 17 is most classifier-sensitive (rank 17 under LR, rank 8 under zero-shot).

Distance-functional robustness. Most distribution-aware metrics correlate only moderately with the adjusted semantic gap (sliced Wasserstein $\rho = 0.525$, Chamfer $\rho = 0.485$), while Gaussian-2-Wasserstein and squared MMD are weaker ($\rho \approx 0.24$–0.26) and C2ST and Grassmann show negative agreement ($\rho = -0.164$ and -0.243; Appendix D). The adjusted gap is primarily a mean-direction finding; shape carries a related but distinct signal.

Synthesis. Two conclusions survive: SDG 15 is robustly least-divergent (rank 17 across configs); SDG 17 is most classifier-sensitive and most divergent under the adjusted gap, while SDG 3 leads under the raw gap for MPNet–LR. The cancellation replicates across all configs and policy-source families. Pooled OLS confirms the adjusted association (Appendix I).

5 Discussion

The headline result is a cancellation: the raw correlation is null because register and topic offset ($\rho = 0.544$ vs. $\rho = -0.390$; Table 3).

5.1 The cancellation is real, not artefactual

Three potential confounds must be ruled out before the cancellation can be interpreted substantively.

Independence of classification and measurement. The LR classifier and the embedding space are separate instruments: classification uses softmax probabilities; measurement uses cosine distance between L2-normalised centroids.

Cross-configuration replication. The cancellation replicates across 8 of 9 configs (Table 3), including concept retrieval and MLP; the register component is negative in 7 of 9; the raw gap is null in all.

SciBERT compression validates the register interpretation. SciBERT compresses the raw gap from 0.34 to 0.04 (88% reduction) while preserving coverage ($\rho = 0.93$; Table 21). Scientific pretraining (Beltagy, Lo, and Cohan, 2019) fuses the registers because both use scientific conventions, exactly what the decomposition predicts.

5.2 Two communities, two dimensions

If the cancellation is real, it demands a theoretical explanation. The two-communities theory (Caplan, 1979) provides one. Caplan characterises social scientists and policy makers as living in "separate worlds with different and often conflicting values, different reward systems, and different languages" (Caplan, 1979, p. 459). The gap, he argues, "necessarily involves value and ideological dimensions as well as technical ones" (Caplan, 1979, p. 461). If the two communities are genuinely distinct knowledge cultures, then a near-zero raw correlation is *expected*, not puzzling: the two corpora develop different topical priorities (producing coverage divergence) and different communicative norms (producing register divergence), and the two effects operate in opposite directions on the semantic gap. Caplan himself warned that "it does not follow... that an alliance of social scientists and policy makers is the panacea" (Caplan, 1979, p. 461). The raw null is the expected result of two distinct knowledge cultures coexisting without deep integration.

The adjusted association reveals something further: coverage and framing are *separable dimensions*, not a single "alignment" property. As shown in Results (Section 4.1), SDGs 9 and 17 illustrate this: SDG 9 has the second-largest coverage gap but a moderate semantic gap, while SDG 17 has moderate coverage but the widest adjusted gap. A single alignment score would conflate these two genuinely different patterns. This separability is the paper's core empirical contribution to the SDG-mapping literature, which has treated coverage as the primary output (Armitage, Lorenz, and Mikki, 2020; Kashnitsky et al., 2024). The framework demonstrates that coverage and framing should be reported as independent axes.

The positive adjusted correlation ($\rho = +0.544$) further confirms that the two dimensions are linked through topic: where one community dominates a given SDG's coverage, the framing tends to diverge. The epistemic-communities framework (Haas, 1992) sharpens this reading. Haas argues that "before states can agree on whether and how to deal collectively with a specific problem, they must reach some consensus about the nature and scope of the problem. . . " (Haas, 1992, p. 29). Where research dominates coverage (SDGs 3, 9), the framing reflects academic norms of knowledge production: "problems set and solved in a context governed by the, largely academic, interests of a specific community" (Gibbons et al., 1994, p. 3). Where policy dominates (SDGs 13, 16, 17), the framing reflects institutional norms of coordination: "knowledge carried out in a context of application" (Gibbons et al., 1994, p. 3). The coverage-semantic linkage is the empirical manifestation of two communities framing the same goals differently.

However, Haas also cautions that "given the many examples of different states reacting in different ways to the same consensual evidence provided by specialists, it is unclear how effective consensual knowledge is, as an independent variable, at explaining or predicting state behavior" (Haas, 1992, p. 30). The adjusted association shows that coverage and framing are linked through topic; it does not show that this linkage produces policy uptake.

The directional decomposition (H1b–H1d) reveals an asymmetric structure, as shown in Table 3. H1d (policy coverage) is the strongest consistent predictor, positive in 9 of 9 configurations. The linkage operates through the policy side: where policy heavily covers an SDG, framing divergence is wider, consistent with Haas's observation that epistemic communities occupy institutional niches (Haas, 1992, p. 30) and with the Mode 1/Mode 2 distinction (Gibbons et al., 1994).

This asymmetry is consistent with Cash et al.'s (Cash et al., 2003) observation that salience, credibility, and legitimacy are "tightly coupled" such that efforts to enhance one "normally incur a cost to the others." Where policy attention is concentrated (SDGs 13, 16, 17), the institutional framing that drives salience simultaneously distances the discourse from research norms of credibility, widening the adjusted gap. The mechanism is structural, not communicative: where policy dominates an SDG, the Mode 2 framing prevails and the gap widens.

5.3 The register decomposition: what the raw gap actually measures

For 16 of 17 SDGs, the register component is positive (+0.076 to +0.261; Table 2), meaning register uniformly inflates the apparent divergence. The implication is stark: any SDG-mapping study that uses raw embedding distances without register adjustment will overestimate semantic divergence. The raw gap is a composite of topic, register, and convention; only the adjusted gap isolates topic.

The exception is SDG 17, where the register component is negative (-0.149): register makes the two communities appear *more similar* than their topics warrant. Under the raw gap SDG 17 ranks near the bottom (0.177); after register removal it jumps to the top (0.326). Policy partnership discourse uses institutional language (*countries*, *nations*, *global*) that superficially resembles research vocabulary, but the substance differs: research concerns technical terms (*model*, *method*), while policy concerns institutional coordination.

SDG 17 is where framing power operates most invisibly: partnership language *sounds* like either community, but the coordination logic belongs exclusively to policy. LeBlanc (Le Blanc, 2015) and ICSU (D. J. Griggs et al., 2017) describe SDG 17 as "implementation infrastructure rather than substantive domains"; the register inversion confirms this empirically.

The register decomposition also reveals what the embedding instrument actually measures. Embedding distance is a composite signal of topic and register (Harris, 1954). The INLP procedure with SDG stratification (Ravfogel et al., 2020) exploits the structural fact that the 17 SDGs supply a near-orthogonal topic set that cross-cuts both registers, allowing linear separation. The method's applicability is bounded by the ontology: in domains where topic and register are tightly confounded (e.g., clinical notes versus biomedical literature), the projection would leak topic into the register direction.

5.4 Implications: topical mention is not alignment

Coverage, the Level 1 metric existing studies report, captures only a weak form of salience; it does not address credibility or legitimacy (Cash et al., 2003). The adjusted gap is a Level 2 metric that begins to measure framing compatibility.

The high-gap SDGs (17, 3, 16, 9, 13) are candidates for boundary-organising practices (Guston, 2001): policy synthesis briefs, joint workshops, or funder-required summaries. Cash et al. (Cash et al., 2003) caution that translation alone cannot resolve the tradeoffs among salience, credibility, and legitimacy. Existing SDG-mapping studies report only coverage (Armitage, Lorenz, and Mikki, 2020); this paper shows coverage and framing are separable dimensions (Section 4.3).

5.5 Limitations

Semantic proximity, not substantive alignment. The study is positioned at Level 2 of a three-level construct (§2.2). Level 3 requires qualitative, institutional, and deliberative methods and is left to future work.

Distance metric discards distributional shape. Centroids are mean-direction measures. Under raw embeddings, distribution-aware metrics (sliced Wasserstein, Chamfer) correlated strongly with the semantic gap ($\rho = 0.44$–0.91), but this agreement was largely a register artifact: the distribution-aware metrics track the register component of the gap, not the adjusted semantic gap (Appendix D). After register removal, the adjusted semantic gap is a mean-direction measure, and distribution-aware metrics correlate with it only weakly to moderately (e.g. Gaussian-2-Wasserstein $\rho = 0.243$, sliced Wasserstein $\rho = 0.525$); they remain an informative full-distribution complement, not a re-ranking. The covariance-shape term accounts for $\approx 79\%$ of the squared Gaussian-2-Wasserstein distance, and a replicate run under the adjacent seed changes the largest per-SDG gap by at most 0.008 (Appendix D).

Corpus scope and asymmetric construction. The research corpus is AI $\cap$ SDG (tight intersection); the policy corpus is broader institutional SDG discourse. The comparison is inherently asymmetric because "AI and the SDGs" is not yet a mainstream organising category in policy discourse. The curated AI/SDG policy family (94 documents) is underpowered for the symmetric check. This was addressed by re-testing the interaction on that family (Section 4.3, Appendix H.6); broadening the research side to all-SDG research would complete the symmetry and is left to future work.

Retrieval-method dependency. The keyword-retrieved corpus trades recall for precision; Appendix A shows that concept-retrieved papers (field-of-study membership) produce closely aligned gap rankings, with a substantially lower SDG 4 share (-7.3pp) as the largest deviation between the two profiles. Re-running the full pipeline with concept-retrieved papers as the primary corpus is the most direct path to eliminating this artefact and is left for future work.

Register adjustment is partial. INLP removes a linear subspace that captures the dominant corpus-level register difference, but register is multi-layered (Biber, 1988). The register interpretation rests on a structural argument (SDG stratification ensures topic cannot leak into the removed direction; §3.8) plus partial empirical validation against independent linguistic markers (Appendix G.2). The validation finds substantial reduction of the corpus-level signal but not its complete elimination.

Per-SDG assignment is approximate. The hard argmax treats SDG discourse as cleanly separable, but the PCA landscape (Figure 4) shows real-world SDG discourse is fuzzy. SDG 4 is artefact-affected by the machine-learning/education "learning" overlap; its rank is read conditional on Appendix B.1. Leave-one-out of SDG 4 preserves the adjusted association ($\rho = 0.597$, $p = 0.016$).

Cross-sectional design. The study observes one period (2018–2025) against a similarly timed policy corpus; it cannot assess whether alignment is improving or deteriorating, or whether specific events (e.g., the EU AI Act 2024, IPCC AR6 2022) (European Union, 2024; IPCC, 2022) shifted research framing.

5.6 The framework as the main contribution

The primary contribution of this paper is the measurement framework, not the specific AI–SDG findings. The framework advances AI–SDG mapping from Level 1 (lexical correspondence) to Level 2 (observed semantic proximity), with explicit marking of Level 3 boundaries (§2.2). The cross-sensitivity transparency (Table 17) is a reusable template: instead of reporting a single gap ranking, it makes the stability of every position transparent across the measurement choices analysts control. The SDG-stratified INLP adaptation — training the corpus classifier on balanced topic–corpus strata so that no linear direction can encode both topic and register — is a methodological contribution applicable to any cross-corpus comparison where topic and register are confounded. The protocol is portable to any research-policy interface that shares a category system and embeds both sides in a common semantic space.

The AI–SDG findings are a first application to be checked against a larger, more diverse policy corpus, not settled results. The findings reported here, the cancellation, the SDG 17 inversion, and the positive adjusted association, are properties of this specific pair of corpora at this moment in time.

6 Conclusion

The Introduction asked whether shared SDG labels between academic AI research and institutional policy discourse indicate genuine alignment. Not without examining framing. This paper demonstrated that coverage divergence (which SDGs each community prioritises) and within-SDG semantic framing divergence (how differently they discuss the same goal) are separable dimensions that can diverge independently. The raw correlation between the two dimensions is near-zero (Spearman $\rho = -0.012$), but this masks a register-topic cancellation: after INLP register removal, the adjusted semantic gap is positively associated with the coverage gap ($\rho = 0.544$, $p = 0.025$). The linkage is asymmetric: policy coverage is the strongest directional predictor of framing divergence (positive in 9/9 configurations), while research coverage shows no consistent signal. Where policy attention concentrates, Mode 2 knowledge production (Gibbons et al., 1994) shapes the discourse in contexts of application, producing framings that are structurally further from research norms; this is the expected consequence of two distinct knowledge cultures producing different framings of the same goals (Caplan, 1979; Cash et al., 2003).

The practical implication is that topical mention is not alignment. SDG tags, badges, and publication counts capture only whether a text is *about* a goal, not how it frames it. The register decomposition reveals that embedding distance, like keyword matching, is a composite signal: for 16 of 17 SDGs, register uniformly inflates the apparent divergence between research and policy. The single exception, SDG 17, is the most instructive case: policy partnership discourse uses institutional language that superficially resembles research vocabulary, masking the deepest topic divergence in the framework. Reporting both dimensions, and acknowledging the register component, is necessary to read the gap correctly. Because funders and policymakers treat SDG tags as evidence of alignment (Heilinger, Kempt, and Nagel, 2024; Heras-Saizarbitoria, Urbieta, and Boiral, 2022), mistaking coverage for framing misreads how SDG engagement is evaluated.

The framework is the paper's primary contribution, advancing AI–SDG mapping from Level 1 (lexical correspondence) to Level 2 (observed semantic proximity) with explicit marking of Level 3 boundaries, and portable to any research-policy interface that shares a category system (Section 5.6). Level 3, whether the divergences documented here reflect productive exploration, institutional failure, or simply different knowledge cultures pursuing different mandates, requires qualitative, institutional, and deliberative methods that lie beyond the scope of embedding analysis.

Reference list

Abadie, Alberto, Susan Athey, Guido W. Imbens, and Jeffrey M. Wooldridge (2020). "Sampling-Based versus Design-Based Uncertainty in Regression Analysis". In: *Econometrica* 88.1, pp. 265–296. DOI: 10.3982/ECTA12675.

Alain, Guillaume and Yoshua Bengio (2018). *Understanding intermediate layers using linear classifier probes*. arXiv: 1610.01644 [stat.ML]. URL: https://arxiv.org/abs/1610.01644.

Armitage, Caroline S., Madeleine Lorenz, and Susanne Mikki (2020). "Mapping scholarly publications related to the Sustainable Development Goals: Do independent bibliometric approaches get the same results?" In: *Quantitative Science Studies* 1.3, pp. 1092–1108. DOI: 10.1162/qss_a_00071.

Babićc, Filippe, Enrique Guerra, Martinćić-Ipśić, and Meśtrovićc (2020). "A Comparison of Approaches for Measuring the Semantic Similarity of Short Texts Based on Word

Embeddings". In: *Journal of Information and Organizational Sciences* 44.2, pp. 231–246. DOI: 10.31341/jios.44.2.2.

Bautista-Puig, Núria, Ana Marta Aleixo, Susana Leal, Ulisses Azeiteiro, and Rodrigo Costas (2021). "Unveiling the Research Landscape of Sustainable Development Goals and Their Inclusion in Higher Education Institutions and Research Centers: Major Trends in 2000–2017". In: *Frontiers in Sustainability* 2, p. 620743. DOI: 10.3389/frsus.2021.620743.

Beltagy, Iz, Kyle Lo, and Arman Cohan (2019). "SciBERT: A Pretrained Language Model for Scientific Text". In: *arXiv preprint arXiv:1903.10676*. DOI: 10.48550/arXiv.1903.10676. arXiv: 1903.10676 [cs.CL]. URL: https://arxiv.org/abs/1903.10676.

Berk, Richard A., Bruce Western, and Robert E. Weiss (1995). "Statistical Inference for Apparent Populations". In: *Sociological Methodology* 25, pp. 421–458. DOI: 10.2307/271073.

Bexell, Magdalena and Kristina Jönsson (2017). "Responsibility and the United Nations' Sustainable Development Goals". In: *Forum for Development Studies* 44.1, pp. 13–29. DOI: 10.1080/08039410.2016.1252424.

Biber, Douglas (1988). *Variation across Speech and Writing*. Cambridge: Cambridge University Press.

Biber, Douglas and Susan Conrad (2009). *Register, Genre, and Style*. Cambridge: Cambridge University Press.

Bornmann, Lutz, Robin Haunschild, and Werner Marx (2016). "Policy documents as sources for measuring societal impact: How often is climate change research mentioned in policy-related documents?" In: *Scientometrics* 109.3, pp. 1477–1495. DOI: 10.1007/s11192-016-2115-y.

Brück, Tim vor der and Marc Pouly (2019). "Text Similarity Estimation Based on Word Embeddings and Matrix Norms for Targeted Marketing". In: *Proceedings of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pp. 1827–1836. DOI: 10.18653/v1/N19-1181.

Caplan, Nathan (1979). "The Two-Communities Theory and Knowledge Utilization". In: *American Behavioral Scientist* 22.3, pp. 459–470. DOI: 10.1177/000276427902200308.

Cash, David W., William C. Clark, Frank Alcock, Nancy M. Dickson, Noelle Eckley, David H. Guston, Jill Jäger, and Ronald B. Mitchell (2003). "Knowledge Systems for Sustainable Development". In: *Proceedings of the National Academy of Sciences* 100.14, pp. 8086–8091. DOI: 10.1073/pnas.1231332100.

Conneau, Alexis, German Kruszewski, Guillaume Lample, Loïc Barrault, and Marco Baroni (2018). *What you can cram into a single vector: Probing sentence embeddings for linguistic properties*. arXiv: 1805.01070 [cs.CL]. URL: https://arxiv.org/abs/1805.01070.

Cover, T. M. and P. E. Hart (1967). "Nearest Neighbor Pattern Classification". In: *IEEE Transactions on Information Theory* 13.1, pp. 21–27. DOI: 10.1109/TIT.1967.1053964.

Cowls, Josh, Andreas Tsamados, Mariarosaria Taddeo, and Luciano Floridi (2021). "A definition, benchmark and database of AI for social good initiatives". In: *Nature Machine Intelligence* 3.2, pp. 111–115. DOI: 10.1038/s42256-021-00296-0.

Cross, Mai'a K. Davis (2013). "Rethinking Epistemic Communities Twenty Years Later". In: *Review of International Studies* 39.1, pp. 137–160. DOI: 10.1017/S0260210512000034.

European Union (2024). *Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*. European Union. URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689.

Fukuda-Parr, Sakiko (2016). "From the Millennium Development Goals to the Sustainable Development Goals: Shifts in Purpose, Concept, and Politics of Global Goal Setting for Development". In: *Gender & Development* 24.1, pp. 43–52. DOI: 10.1080/13552074.2016.1145895.

Gibbons, Michael, Camille Limoges, Helga Nowotny, Simon Schwartzman, Peter Scott, and Martin Trow (1994). *The New Production of Knowledge: The Dynamics of Science and Research in Contemporary Societies*. London: SAGE Publications.

Gjorgjevikj, Ana, Kostadin Mishev, Dimitar Trajanov, and Ljupco Kocarev (2025). "Benchmarking Sentence Encoders in Associating Indicators With Sustainable Development Goals and Targets". In: *IEEE Access* 13, pp. 141434–141460. DOI: 10.1109/ACCESS.2025.3595894.

Gohr, C., G. Rodríguez, S. Belomestnykh, D. Berg-Moelleken, N. Chauhan, J.-O. Engler, L. V. von Heydebreck, M. J. Hintz, M. Kretschmer, C. Krügermeier, J. Meinberg, A.-L. Rau, C. Schwenck, I. Aoulkadi, S. Poll, E. Frank, F. Creutzig, O. Lemke, M. Maushart, J. Pfendtner-Heise, J. Rathgens, and H. von Wehrden (2025). "Artificial intelligence in sustainable development research". In: *Nature Sustainability* 8, pp. 970–978. DOI: 10.1038/s41893-025-01598-6.

Grieve, Jack, Sara Bartl, Matteo Fuoli, Jason Grafmiller, Weihang Huang, Alejandro Jawerbaum, Akira Murakami, Marcus Perlman, Dafne Roemling, and Bodo Winter (2025). "The Sociolinguistic Foundations of Language Modeling". In: *Frontiers in Artificial Intelligence* 7, p. 1472411. DOI: 10.3389/frai.2024.1472411.

Griggs, D. J., M. Nilsson, A. Stevance, and D. McCollum, eds. (2017). *A Guide to SDG Interactions: From Science to Implementation*. International Council for Science. DOI: 10.24948/2017.01.

Guston, David H. (2001). "Boundary Organizations in Environmental Policy and Science: An Introduction". In: *Science, Technology, & Human Values* 26.4, pp. 399–408. DOI: 10.1177/016224390102600401.

Haas, Peter M. (1992). "Introduction: Epistemic Communities and International Policy Coordination". In: *International Organization* 46.1, pp. 1–35. DOI: 10.1017/S0020818300001442.

Harris, Zellig S. (1954). "Distributional Structure". In: *WORD* 10.2–3, pp. 146–162. DOI: 10.1080/00437956.1954.11659520.

Hastie, Trevor, Robert Tibshirani, and Jerome Friedman (2009). *The Elements of Statistical Learning*. 2nd. Springer. DOI: 10.1007/978-0-387-84858-7.

Heilinger, Jan-Christoph, Hendrik Kempt, and Saskia K. Nagel (2024). "Beware of sustainable AI! Uses and abuses of a worthy goal". In: *AI and Ethics* 4.2, pp. 201–212. DOI: 10.1007/s43681-023-00259-8.

Hellwig, Joerg, Sarah Huggett, and Mark Siebert (2019). *Artificial Intelligence: How knowledge is created, transferred, and used*. Elsevier.

Heras-Saizarbitoria, Iñaki, Laida Urbieta, and Olivier Boiral (2022). "Organizations' Engagement with Sustainable Development Goals: From Cherry-Picking to SDG-Washing?" In: *Corporate Social Responsibility and Environmental Management* 29.2, pp. 316–328. DOI: 10.1002/csr.2202.

Hickel, Jason (2019). "The Contradiction of the Sustainable Development Goals: Growth versus Ecology on a Finite Planet". In: *Sustainable Development* 27.5, pp. 873–884. DOI: 10.1002/sd.1947.

IPCC (2022). *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report*. Intergovernmental Panel on Climate Change. URL: https://www.ipcc.ch/report/ar6/wg2/.

Jankin, Slava, Alexander Baturo, and Niheer Dasandi (2024). "Words to Unite Nations: The Complete United Nations General Debate Corpus, 1946–present". In: *Journal of Peace Research* 62.4, pp. 1339–1351. DOI: 10.1177/00223433241275335.

Kashnitsky, Yury, Guillaume Roberge, Jingwen Mu, Kevin Kang, Weiwei Wang, Maurice Vanderfeesten, Maxime Rivest, Savvas Chamezopoulos, Robert Jaworek, Maéva Vignes, Bamini Jayabalasingham, Finne Boonen, Chris James, Marius Doornenbal, and Isabelle Labrosse (2024). "Evaluating Approaches to Identifying Research Supporting the United

Nations Sustainable Development Goals". In: *Quantitative Science Studies* 5.2, pp. 408–425. DOI: 10.1162/qss_a_00304.

Krizhevsky, Alex, Ilya Sutskever, and Geoffrey E Hinton (2012). "ImageNet Classification with Deep Convolutional Neural Networks". In: *Advances in Neural Information Processing Systems*. Ed. by F. Pereira, C.J. Burges, L. Bottou, and K. Weinberger. Vol. 25. Curran Associates, Inc. URL: https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf.

Le Blanc, David (2015). "Towards Integration at Last? The Sustainable Development Goals as a Network of Targets". In: *Sustainable Development* 23.3, pp. 176–187. DOI: 10.1002/sd.1582.

Manning, Christopher D., Prabhakar Raghavan, and Hinrich Schütze (2008). *Introduction to Information Retrieval*. Cambridge University Press. URL: https://nlp.stanford.edu/IR-book/.

McCarthy, Chris, Cassandra Brooks, Troy Sternberg, Kyle Shaney, and Buho Hoshino (2026). "AI-Assisted Multi-Target Classification for Research-Policy Alignment in Conservation Science". In: *Ecological Informatics* 94, p. 103669. DOI: 10.1016/j.ecoinf.2026.103669.

Nedungadi, Prema, Sreejith Surendran, Kai-Yu Tang, and Raghu Raman (2024). "Big data and AI algorithms for sustainable development goals: a topic modeling analysis". In: *IEEE Access* 12, pp. 188519–188541. DOI: 10.1109/ACCESS.2024.3516500.

Nilsson, Måns, Dave Griggs, and Martin Visbeck (2016). "Policy: Map the Interactions between Sustainable Development Goals". In: *Nature* 534.7607, pp. 320–322. DOI: 10.1038/534320a.

Niu, Xing and Marine Carpuat (Sept. 2017). "Discovering Stylistic Variations in Distributional Vector Space Models via Lexical Paraphrases". In: *Proceedings of the Workshop on Stylistic Variation*. Ed. by Julian Brooke, Thamar Solorio, and Moshe Koppel. Copenhagen, Denmark: Association for Computational Linguistics, pp. 20–27. DOI: 10.18653/v1/W17-4903. URL: https://aclanthology.org/W17-4903/.

OECD (2024). *OECD.AI: Partner Data and Methodological Note*. OECD.AI Observatory methodological documentation. URL: https://oecd.ai/en/partner-data-methodological-note.

Priem, Jason, Heather Piwowar, and Richard Orr (2022). *OpenAlex: A fully-open index of the world's scholarly works*. DOI: 10.48550/arXiv.2205.01833. arXiv: 2205.01833.

Pukelis, Lukas, Núria Bautista-Puig, Gustė Statulevičiūtė, Vilius Stančiauskas, Gokhan Dikmener, and Dina Akylbekova (2022). "OSDG 2.0: a multilingual tool for classifying text data by UN Sustainable Development Goals (SDGs)". In: *arXiv preprint*. DOI: 10.48550/arXiv.2211.11252. eprint: 2211.11252.

Ravfogel, Shauli, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg (2020). "Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection". In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, pp. 7237–7256. DOI: 10.18653/v1/2020.acl-main.647.

Reimers, Nils and Iryna Gurevych (2019). "Sentence-BERT: Sentence embeddings using Siamese BERT-networks". In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, pp. 3982–3992. DOI: 10.18653/v1/D19-1410.

Rodriguez, Pedro L., Arthur Spirling, and Brandon M. Stewart (2023). "Embedding Regression: Models for Context-Specific Description and Inference". In: *American Political Science Review* 117.4, pp. 1255–1274. DOI: 10.1017/S0003055422001228.

Schoeberl, M., B. Toney, and M. Dunham (2023). *Identifying AI Research*. Tech. rep. Center for Security and Emerging Technology (CSET). DOI: 10.51593/20230.

SDG Classification Expert Group (2025). *SDG Classification Benchmark*. Expert-labelled benchmark for SDG text classification, all 17 SDGs. URL: https://github.com/SDGClassification/benchmark.

SDGi (2024). *SDGi Corpus of Voluntary National and Local Reviews*. Available on Hugging Face. URL: https://huggingface.co/datasets/UNDP/sdgi-corpus.

Sentence-Transformers (2021a). *all-MiniLM-L6-v2: Sentence Embedding Model*. https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2.

— (2021b). *all-mpnet-base-v2: Sentence Embedding Model*. https://huggingface.co/sentence-transformers/all-mpnet-base-v2.

Singh, Aakash, Anurag Kanaujia, Vivek Kumar Singh, and Ricardo Vinuesa (2024). "Artificial intelligence for Sustainable Development Goals: Bibliometric patterns and concept evolution trajectories". In: *Sustainable Development* 32.1, pp. 724–754. DOI: 10.1002/sd.2706.

Sioumalas-Christodoulou, Konstantinos and Aristides Tympas (2025). "AI metrics and policymaking: assumptions and challenges in the shaping of AI". In: *AI & Society* 40, pp. 4655–4670. DOI: 10.1007/s00146-025-02181-5.

Spearman, Charles (1904). "The Proof and Measurement of Association between Two Things". In: *The American Journal of Psychology* 15.1, pp. 72–101. DOI: 10.2307/1412159.

Strauss, Ilan, Isobel Moure, Tim O'Reilly, and Sruly Rosenblat (Apr. 2025). *Real-World Gaps in AI Governance Research: AI Safety and Reliability in Everyday Deployments.* DOI: 10.48550/arXiv.2505.00174. arXiv: 2505.00174.

Toney-Wails, Autumn, Kathleen Curlee, and Emelia Probasco (2024). "Trust Issues: Discrepancies in Trustworthy AI Keywords Use in Policy and Research". In: *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT '24).* Association for Computing Machinery, pp. 2222–2233. DOI: 10.1145/3630106.3659035.

Vanderfeesten, Maurice, Eike Spielberg, and Yassin Gunes (2020). *Survey Data of "Mapping Research Output to the Sustainable Development Goals (SDGs)" by Aurora Universities Network (AUR).* Dataset, version 1.0.1. DOI: 10.5281/zenodo.3798385.

Vinuesa, Ricardo, Hossein Azizpour, Iolanda Leite, Madeline Balaam, Virginia Dignum, Sami Domisch, Anna Felländer, Simone Daniela Langhans, Max Tegmark, and Francesco Fuso Nerini (2020). "The role of artificial intelligence in achieving the Sustainable Development Goals". In: *Nature Communications* 11.1, p. 233. DOI: 10.1038/s41467-019-14108-y.

Wang, Kuansan, Zhihong Shen, Chiyuan Huang, Chieh-Han Wu, Yuxiao Dong, and Anshul Kanakia (2020). "Microsoft Academic Graph: When experts are not enough". In: *Quantitative Science Studies* 1.1, pp. 396–413. DOI: 10.1162/qss_a_00021.

Wulff, Dirk U., Dominik S. Meier, and Rui Mata (2023). *SDG Knowledge Hub Dataset of SDG-labeled News Articles.* IISD SDG Knowledge Hub; CC-BY 4.0; 9,172 articles. DOI: 10.5281/zenodo.7523032.

Yin, Hui, Amir Aryani, Gavin Lambert, Zhuochen Wu, Nakul Nambiar, Marcus White, Luis Salvador-Carulla, Shazia Sadiq, Elvira Sojli, Jennifer Boddy, Greg Murray, and Wing Wah Tham (2025). "Leveraging artificial intelligence technology for mapping publications to Sustainable Development Goals". In: *Array* 27, p. 100419. DOI: 10.1016/j.array.2025.100419.

A Retrieval and Query Chain

The research corpus was retrieved from the OpenAlex API (Priem, Piwowar, and Orr, 2022) using 68 queries formed by the cross-product of OpenAlex's native SDG work filter and four AI search terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`), restricted to `publication_year > 2017`. The 2018 start date reflects the post-AlexNet (Krizhevsky, Sutskever, and Hinton, 2012) emergence of deep learning as the dominant paradigm and a research-uptake lag after the SDGs' adoption in 2015. OpenAlex SDG metadata defines the candidate universe; the SDG assignments used in the analysis are produced by the independently validated supervised classifier (Section 3.5). Complete query construction and credential handling are implemented in `1_code/0_fetch/fetch_openalex.py`.

The retrieval boundary is a deliberate scope condition. Elsevier's data-driven characterisation of AI research draws on over 700 field-specific keywords across some 600,000 documents (Hellwig, Huggett, and Siebert, 2019), favouring recall. The OECD.AI Observatory methodology instead defines a paper as AI when it carries the Artificial Intelligence or Machine Learning field of study in the Microsoft Academic Graph (MAG), whose Fields of Study taxonomy OpenAlex later inherited and extended by mapping it to Wikidata identifiers (OECD, 2024; Priem, Piwowar, and Orr, 2022; Wang et al., 2020). The present design prioritises precision: a four-term query keeps non-AI noise out of the embedding space at the cost of recall in specialised AI subfields. Schoeberl, Toney, and Dunham (2023) reach the same conclusion for AI-research identification, namely that results are sensitive to the retrieval method and that supervised classification is preferred.

A direct test of this scope condition re-ran retrieval using field-of-study membership for the same two AI fields, mirroring the OECD.AI definition (OECD, 2024), and re-scored the resulting corpus through the identical pipeline. No SDG filter was applied, and SDG assignment used the same trained logistic-regression classifier used throughout (Section 3.5), so the only difference from the main corpus is the retrieval strategy. Concept retrieval used OpenAlex concept tags for Artificial Intelligence and Machine Learning, capped at `MAX_PAPERS_CONCEPT_CORPUS` = 100,000 papers, yielding the roughly 100,000-paper concept corpus. This scale sits on the empirical variance plateau established in Appendix E: at 50,000 papers the coverage-profile standard deviation is 0.001 and the mean within-SDG semantic gap is 0.342, within 0.002 of the full-corpus estimate of 0.340 (full corpus N = 3,105,144), while policy-text calibration bias has also converged (0.122). The scale also exceeds every corpus size used in prior SDG classification and mapping benchmarks, namely the OSDG community dataset (30,532 single-label texts) Pukelis et al. (2022), the SDG benchmark encoder evaluation corpus (35,001 text excerpts) Gjorgjevikj et al. (2025), the SDGi corpus (20,267 single-label segments) SDGi (2024), the Aurora expert-validated set (4,971 single-label texts) Vanderfeesten, Spielberg, and Gunes (2020), and the 6,000-paper benchmark by Kashnitsky et al. (2024), while remaining far smaller than the full 3,105,144-segment research corpus and thus a computationally tractable substrate for OpenAlex concept tagging.

Table 4 compares the SDG coverage shares of the keyword-retrieved and concept-retrieved corpora. The two profiles are closely aligned (Spearman $\rho = 0.948$, Kendall $\tau = 0.868$); the largest deviations are SDG 4 (-7.3pp, the known ML-vocabulary artefact, Appendix B.1) and SDG 9 (+5.8pp). The within-SDG coverage-gap and semantic-gap rankings are also stable against the keyword baseline (coverage-gap rank $\rho = 0.87$; semantic-gap rank $\rho = 0.81$, Table 17). We therefore report the main findings as robust to the query-term choice, while noting that both corpora remain conditional on their respective retrieval strategies.

Table 4. SDG coverage-share comparison: keyword-retrieved versus concept-retrieved (AI/ML field-of-study) research corpus.

SDG	Keyword %	Concept %	Δ
1	0.4	0.7	+0.3
2	2.4	1.8	-0.6
3	25.7	25.7	-0.0
4	15.6	8.3	-7.3
5	2.0	1.7	-0.2
6	1.0	0.8	-0.2
7	4.1	4.7	+0.6
8	2.2	2.3	+0.1
9	20.1	25.9	+5.8
10	0.9	0.8	-0.1
11	5.1	8.0	+2.9
12	1.5	2.7	+1.2
13	3.7	4.2	+0.5
14	1.6	1.3	-0.3
15	2.2	2.1	-0.1
16	11.2	8.8	-2.4
17	0.3	0.2	-0.0
Spearman $\rho = 0.948$; Kendall $\tau = 0.868$			

Notes: Δ is concept minus keyword share in percentage points.

B Diagnostics for Reference Classifier

B.1 SDG 4 Lexical Artefact Audit: The ML 'Learning' Overlap

SDG 4 is the only goal audited because it is the only one carrying an a priori lexical-overlap concern: "learning" is shared ML and education vocabulary, and prior AI–SDG inventories place SDG 4 outside the leading tier of SDG 3 and SDG 7 (Singh et al., 2024; Cowls et al., 2021; Nedungadi et al., 2024), so its third-place research share here is suspect on independent grounds. The audit does not relabel records; it tests whether the concern is empirically plausible. Table 5 therefore compares the lexical composition of SDG 4-assigned research with a matched non-SDG4 sample and with SDG 9, using only two transparent keyword groups: ML-learning and education-specific vocabulary.

The result (Table 5) is a negative test of the false-positive hypothesis: education-specific vocabulary alone covers 44.4% of SDG 4-assigned records and overlaps ML vocabulary in a further 23.8%, while only 11.8% are ML-only. The comparison sets are inverted, with ML-only vocabulary dominating the non-SDG4 sample and the SDG 9 set (50.3% and 61.5%). The SDG 4 signal is therefore not the same technical language that drives SDG 9; but the 23.8% "both" share (versus 2.6% and 2.9% elsewhere) shows the two vocabularies do overlap materially within the SDG 4-assigned set.

The main analysis therefore retains the canonical LR argmax assignment but continues to treat SDG 4 as artefact-affected. The most accurate reading is a learning-related assignment whose interpretation is blurred by lexical overlap: neither "clean education" nor "pure ML artefact."

Table 5. SDG 4 lexical artefact audit.

Corpus subset	N	ML-only %	Education-only %	Both %	Neither / unclear %
SDG 4-assigned research	470002	11.8	44.4	23.8	19.9
Non-SDG4 research sample	470002	50.3	4.0	2.6	43.1
SDG 9-assigned research	577833	61.5	2.5	2.9	33.1

Notes: The audit only checks whether SDG 4-assigned research is disproportionately dominated by machine-learning vocabulary, education vocabulary, or both.

B.2 Centroid Similarity

This subsection reports the pairwise cosine-similarity heatmap (Figure 7) for the 17 canonical SDG centroids. The centroids are the per-SDG mean training embeddings (L2-normalised) used by the LR classifier.

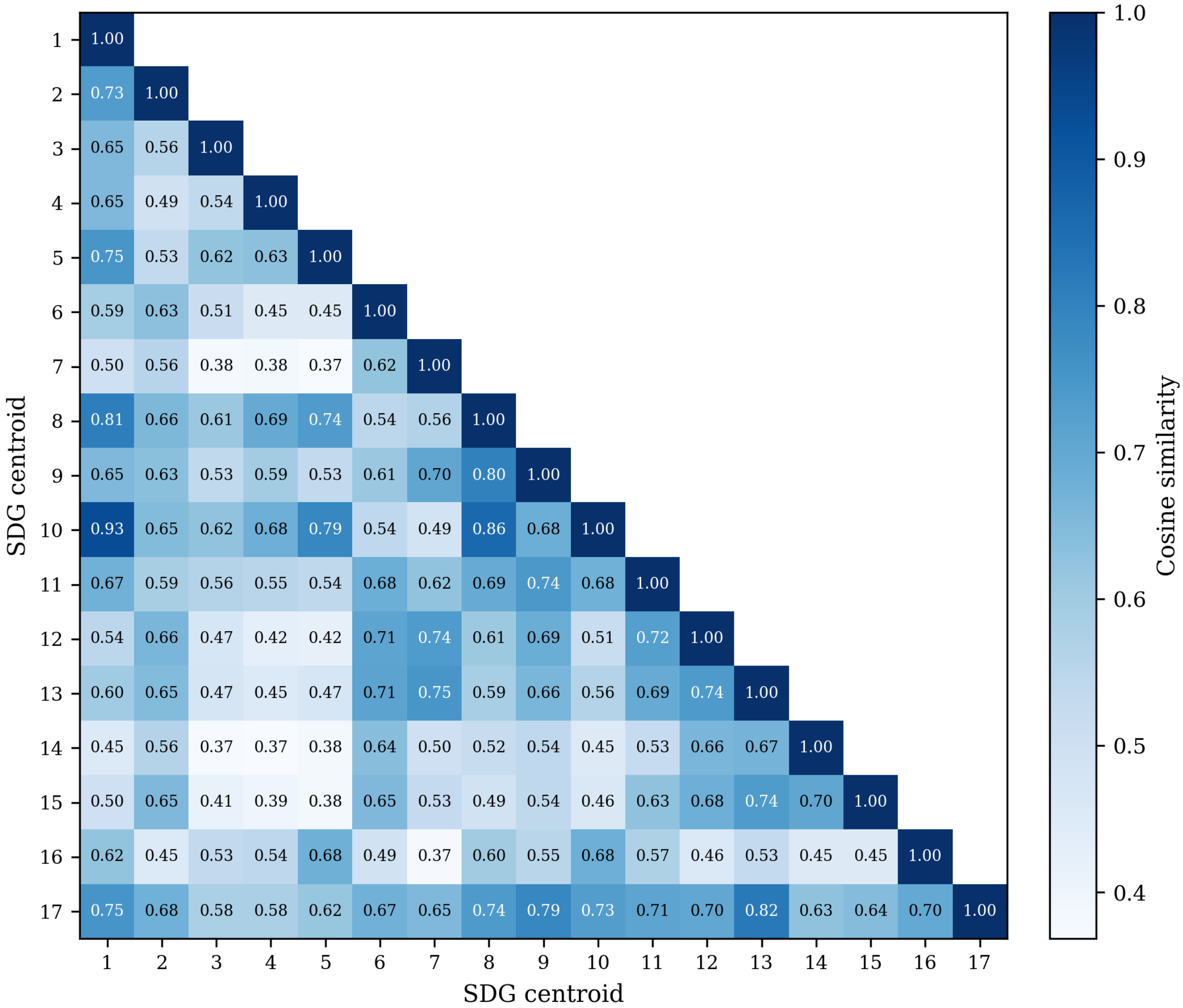

Figure 7. Pairwise cosine similarity between canonical SDG centroids (lower triangle). SDG 13 and SDG 17 sit close together (cosine 0.821), as do SDGs 1, 8, and 10.

Figure 7 shows the full 17×17 similarity grid. Reading the grid, three observations stand out. First, the diagonal dominates: most centroids are far more similar to themselves than to any neighbour, which is why the supervised LR classifier (Table 1) can build on this well-separated base to reach 0.816. Second, a small set of off-diagonal cells is elevated: SDG 11's centroid neighbours SDG 9 (cosine 0.744), SDG 8 lies near SDGs 1 and 10 (cosine 0.808 and 0.862), and SDGs 1 and 10 are themselves the closest off-diagonal pair (cosine 0.931) – these are the proximities that make the SDG 10 and SDG 8 scores in Section 4.1 indicative rather than exact. Third, SDG 17 is the clear outlier: its row and column are the flattest, and its nearest neighbour is SDG 13 (cosine 0.821), reflecting the overlap between climate and partnership discourse noted in Section 4.1.

C Lexical Illustration of the Semantic Gap

Table 6 lists the TF-IDF terms most distinctive of the research and policy sides for each SDG. Generic ML methodology vocabulary, such as *model*, *learning*, and *neural*, is excluded because these terms reflect the search strategy rather than SDG-specific framing.

The contrasts echo the embedding patterns. The widest-gap SDGs (3, 13, 11, 7, 1, 8, 14; raw gap ≥ 0.358) consistently oppose technical measurement vocabulary, *detection*, *images*, *yield*, *regression*, against institutional or welfare terms, *climate*, *emissions*, *health*, *employment*. SDG 13 clarifies the pattern most sharply: research terms centre on crop monitoring and soil science while policy terms address national emissions commitments. Lower-gap SDGs (17, 15, 4, 10, 5, 6, 2; raw gap ≤ 0.299) show narrower lexical divergence, such as the shared technology-infrastructure frame of SDG 9 or the social-media-partnership overlap of SDG 17. A further observation cuts across both groups: agricultural-AI vocabulary (*crop*, *soil*, *agricultural*) appears on the research side of nine unrelated SDGs, reflecting a strong agri-tech skew in the AI-for-SDG literature rather than alignment with each goal's policy domain.

Table 6. Distinctive TF-IDF terms for all 17 SDGs.

SDG	Gap	Research-side distinctive terms	Policy-side distinctive terms
1	0.415	alleviation, china, approach, zakat, satellite, household, studies, indonesia, literature, relationship	social, cent, national, people, protection, services, population, vulnerable, living, million
2	0.299	crop, detection, plant, diseases, images, leaf, precision, techniques, soil, yield	population, food, children, prevalence, worst, access, hunger, million, births, ratio
3	0.494	patients, cancer, cells, images, clinical, detection, disease, potential, genes, predict	health, services, care, national, population, births, mortality, rate, people, access
4	0.267	cognitive, brain, neurons, sleep, agricultural, food, cells, motor, memory, poor	education, school, secondary, primary, national, quality, children, percent, access, vocational
5	0.278	menstrual, poor, pain, mothers, differences, relationship, patients, fertility, stress, sex	gender, violence, equality, national, rights, girls, men, cent, sexual, domestic
6	0.292	irrigation, soil, agricultural, crop, salinity, agriculture, sup, potential, sub, yield	sanitation, access, drinking, supply, national, wastewater, cent, services, population, clean
7	0.428	agricultural, poverty, agriculture, battery, optimization, approach, load, drying, poor, artificial	access, electricity, renewable, clean, worst best, population, affordable, index worst, cent, national
8	0.387	detection, image, food, recognition, leaf, plant, cnn, convolutional, disease, techniques	growth, employment, gdp, worst, labour, best, economic, cent, government, tourism
9	0.314	agricultural, agriculture, iot, detection, food, crop, plant, farming, farmers, soil	infrastructure, innovation, national, government, public, countries, strategy, access, population, intelligence
10	0.271	wealth, income inequality, effects, evidence, studies, que, distribution, literature, mobility, poor	people, rights, countries, migration, national, cent, disabilities, persons, discrimination, target
11	0.449	detection, image, accident, traffic, object, poor, recognition, spatial, fusion, agricultural	housing, city, urban, transport, cities, public, waste, national, plan, green
12	0.296	food, agricultural, detection, quality, potential, behavior, samples, organic, design, compounds	waste, consumption, national, recycling, production, procurement, management, public, economy, circular
13	0.480	soil, agricultural, crop, moisture, drought, sub, rice, stress, agriculture, yield	climate, change, emissions, countries, national, action, global, energy, nations, united
14	0.358	detection, images, aquaculture, food, bacterial, shrimp, cells, imaging, farming, techniques	marine, fish caught, biodiversity, rate, ocean, worst, population, sea, embodied, coastal
15	0.239	soil, crop, agricultural, plant, spectral, images, sensing, detection, remote, field	biodiversity, forest, worst, national, protected, population, area, freshwater, species, best
16	0.329	poor, food, covid-, pain, image, detection, social, studies, patients, brain	international, nations, peace, rights, united, security, human, law, world, global
17	0.177	sentiment, tweets, covid-, twitter, social, agricultural, dan, yang, zakat, media	countries, oecd, international, nations, united, agenda, report, financing, global, challenges

Notes: Terms are TF-IDF-weighted unigrams and bigrams from deterministic samples of research and policy texts assigned to each SDG.

D Distance-Functional Robustness Table

Table 7 reports the per-SDG research–policy gap under the full battery of distribution-aware metrics, together with the primary semantic gap. Part 1 covers the full-corpus exact metrics (sliced Wasserstein, Chamfer, Hausdorff, Gaussian-2-Wasserstein, Grassmann, linear MMD); part 2 covers the sampled metrics (exact EMD, Sinkhorn, energy, squared RBF MMD, C2ST AUC) plus the sampled semantic gap. The final row of each part reports Spearman ρ between each metric's ranking and the primary (adjusted) ranking.

Why the raw robustness result does not carry over. Under raw embeddings the distribution-aware metrics appeared to preserve the semantic gap ranking ($\rho = 0.44$–0.91), an agreement driven by the shared *register* component rather than by topic. Decomposing the raw gap into adjusted semantic gap + register component, the register component has *larger* variance than the raw gap itself (variance ratio 1.24) and correlates $\rho = 0.67$ with the raw gap while correlating $\rho = -0.29$ with the adjusted semantic gap. The distribution-aware metrics therefore tracked register, agreeing with the raw semantic gap but diverging from the adjusted one. After INLP removes register, the metrics instead track topic-space *shape*: most correlate moderately with the adjusted semantic gap ($\rho = 0.44$–0.69; sliced Wasserstein $\rho = 0.525$, exact EMD $\rho = 0.694$), Gaussian-2-Wasserstein and squared MMD are weaker ($\rho \approx 0.24$–0.26), and Sinkhorn, C2ST, and Grassmann show near-zero or negative agreement ($\rho = 0.047$, -0.164, -0.243), and they elevate SDGs 12 and 1 at the margin. The adjusted semantic gap is a mean-direction estimate, and the distribution-aware distances are a complementary full-distribution view. The covariance-shape term of the Gaussian-2-Wasserstein distance accounts for $\approx 79\%$ of the squared distance on average, and a replicate run under the adjacent seed changes the largest per-SDG gap by at most 0.008.

The H1 decomposition is not an artefact of the centroid estimate. Table 9 re-runs the H1a–H1d correlation grid with the distribution-aware semantic gaps (adjusted embeddings) in place of the centroid gap; predictors and test are identical to Table 3. The positive coverage–framing association replicates under the mean-complementary family: sliced Wasserstein ($\rho = 0.554$, $p = 0.022$), linear MMD ($\rho = 0.603$, $p = 0.012$), and energy distance ($\rho = 0.537$, $p = 0.027$) reproduce the adjusted association at magnitudes close to the centroid estimate ($\rho = 0.544$, $p = 0.025$), with squared RBF MMD directionally consistent ($\rho = 0.473$, $p = 0.056$). The distributional view also surfaces a nuance the mean-direction estimate misses: H1c (research coverage) is positively associated with the sliced-Wasserstein ($\rho = 0.539$, $p = 0.028$), energy-distance ($\rho = 0.532$, $p = 0.030$), and squared RBF MMD ($\rho = 0.551$, $p = 0.024$) gaps even though it is null on the centroid gap, suggesting research-heavy goals show wider distributional shape divergence without a wider mean-direction gap. The association is not uniform across the battery: exact EMD

Table 7. Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 1 of 2: full-corpus exact metrics). SWD = sliced Wasserstein; MMD^2 = squared RBF maximum mean discrepancy; Gauss. W_2 = Gaussian 2-Wasserstein; C2ST = classifier two-sample test AUC. The final row reports Spearman ρ between each metric's ranking and the canonical ranking across the 17 SDGs.

SDG	Canonical	SWD	Chamfer	Hausdorff	Gauss. W_2	Grassmann	linear MMD
1	0.191	0.009	0.509	0.600	0.684	3.727	0.067
2	0.089	0.008	0.388	0.454	0.618	3.629	0.057
3	0.319	0.011	0.454	0.581	0.717	3.717	0.129
4	0.158	0.010	0.399	0.487	0.607	3.307	0.096
5	0.151	0.010	0.429	0.529	0.646	3.514	0.095
6	0.130	0.009	0.417	0.475	0.639	3.598	0.080
7	0.170	0.010	0.427	0.522	0.687	3.729	0.105
8	0.184	0.008	0.474	0.564	0.658	3.566	0.065
9	0.228	0.010	0.430	0.556	0.671	3.444	0.104
10	0.195	0.008	0.456	0.517	0.617	3.343	0.062
11	0.209	0.009	0.446	0.542	0.677	3.484	0.091
12	0.156	0.009	0.467	0.566	0.677	3.462	0.094
13	0.219	0.010	0.417	0.498	0.667	3.667	0.117
14	0.132	0.010	0.415	0.510	0.675	3.753	0.098
15	0.084	0.008	0.394	0.464	0.578	3.421	0.056
16	0.242	0.010	0.417	0.520	0.640	3.333	0.103
17	0.326	0.009	0.461	0.490	0.595	2.993	0.104
ρ vs. canonical	–	0.525	0.485	0.444	0.243	-0.243	0.620

and Chamfer agree moderately with the centroid ranking (Table 7, final row) yet show no H1a association, and every metric whose ranking agreement is weak or negative is null as well, with one exception: Sinkhorn's H1d is significantly negative ($\rho = -0.505$, $p = 0.041$), an outlier consistent with its near-zero ranking agreement. Excluding SDG 4 leaves every conclusion unchanged (full $n = 16$ grid in the analysis JSON).

Table 8. Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 2 of 2: sampled metrics; continuation of Table 7). EMD = exact 1-Wasserstein; MMD^2 = squared RBF-MMD; Centroid (samp.) = sampled centroid-gap control column.

SDG	Canonical	EMD	Sinkhorn	Energy	MMD^2	C2ST AUC	Centroid (samp.)
1	0.191	0.645	0.623	0.056	0.025	0.969	0.186
2	0.089	0.533	0.516	0.054	0.026	0.979	0.088
3	0.319	0.665	0.597	0.105	0.046	0.985	0.315
4	0.158	0.554	0.523	0.084	0.040	0.988	0.157
5	0.151	0.581	0.543	0.082	0.038	0.984	0.152
6	0.130	0.546	0.547	0.073	0.036	0.986	0.128
7	0.170	0.581	0.545	0.094	0.045	0.990	0.170
8	0.184	0.620	0.596	0.054	0.024	0.978	0.185
9	0.228	0.625	0.560	0.090	0.040	0.984	0.228
10	0.195	0.602	0.636	0.051	0.023	0.964	0.191
11	0.209	0.606	0.580	0.076	0.035	0.977	0.213
12	0.156	0.605	0.605	0.081	0.037	0.987	0.154
13	0.219	0.577	0.485	0.102	0.048	0.983	0.217
14	0.132	0.555	0.538	0.091	0.044	0.984	0.134
15	0.084	0.515	0.515	0.053	0.026	0.982	0.083
16	0.242	0.604	0.495	0.088	0.039	0.983	0.241
17	0.326	0.582	0.499	0.083	0.037	0.980	0.319
ρ vs. canonical	–	0.694	0.047	0.444	0.260	-0.164	1.000

Table 9. H1a–H1d coverage-predictor vs semantic-gap Spearman correlations under distribution-aware semantic gaps (adjusted embeddings). Distributional gaps replace the centroid gap as the outcome vector; predictors and test are identical to Table 3.

	H1a Cov. gap	H1b Dominance	H1c Research cov.	H1d Policy cov.
Sliced Wasserstein	$+0.554^{*}$	+0.061	$+0.539^{*}$	+0.397
Linear MMD	$+0.603^{*}$	+0.022	$+0.466^{\dagger}$	$+0.439^{\dagger}$
Energy distance	$+0.537^{*}$	+0.074	$+0.532^{*}$	+0.384
Squared RBF MMD	$+0.473^{\dagger}$	+0.113	$+0.551^{*}$	+0.296
Exact EMD	+0.304	+0.147	+0.167	+0.217
Chamfer	+0.103	-0.115	-0.350	-0.006
Modified Hausdorff	+0.159	+0.152	+0.127	+0.066
Gaussian-2-Wasserstein	-0.007	+0.240	+0.262	-0.042
Sinkhorn	-0.199	+0.397	-0.176	-0.505^{*}
C2ST AUC	+0.270	+0.275	+0.375	-0.139
Grassmann	-0.186	+0.034	+0.012	-0.228

Notes: Spearman ρ with two-sided Monte Carlo permutation p-values (100,000 resamples, seed 42), $n = 17$ SDGs; $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.10$. Excluding SDG 4 ($n = 16$) leaves every conclusion unchanged; the full $n = 16$ grid is in the analysis JSON (distributional_h1_interaction.json).

E Sample-Stability Robustness Check

To test whether the headline findings depend heavily on research-corpus size, this study reran the downstream research-side analysis on repeated random subsamples drawn from the pre-embedded, pre-scored OpenAlex corpus. For each of 11 sampled tiers (from 1,000 to 2,000,000 papers), 100 independent samples were drawn without replacement, the research coverage profile was recomputed, sampled research centroids were rebuilt, the mean within-SDG semantic gap was recalculated, and the policy-text calibration-bias summary was re-estimated. Policy-side quantities were held fixed.

Table 10. Sample-stability ladder for the research corpus.

Sample Size	Mean SD of SDG coverage shares	Mean within-SDG semantic gap	Policy-text calibration bias
1,000	0.006	0.417 ± 0.020	0.122 ± 0.008
2,000	0.004	0.384 ± 0.015	0.122 ± 0.006
5,000	0.003	0.358 ± 0.009	0.122 ± 0.004
10,000	0.002	0.350 ± 0.006	0.122 ± 0.002
20,000	0.001	0.344 ± 0.005	0.122 ± 0.002
50,000	0.001	0.342 ± 0.003	0.122 ± 0.001
100,000	0.001	0.341 ± 0.003	0.122 ± 0.001
200,000	0.000	0.340 ± 0.002	0.122 ± 0.001
500,000	0.000	0.340 ± 0.001	0.122 ± 0.000
1,000,000	0.000	0.340 ± 0.001	0.122 ± 0.000
2,000,000	0.000	0.340 ± 0.000	0.122 ± 0.000
Full corpus	–	0.340	0.122

Notes: Mean SD of SDG coverage shares is the unweighted mean of the per-SDG standard deviations in research coverage share across the 100 random draws at each sampled tier. The remaining columns report the mean and draw-to-draw standard deviation of the headline downstream metrics. The full-corpus row is deterministic and therefore has no variance term.

Plain-language guide.[2] The stability ladder shows that the clearest practical plateau appears by roughly 200,000 papers, where the mean semantic gap (∼0.340) and policy-text calibration bias (∼0.122) already stabilise to within 0.001 of their full-corpus values. From 500,000 papers upward, draw-to-draw dispersion is negligible. The full 3,105,144-segment result is retained as the primary estimate because it uses all available data, yields the tightest values, and enables per-SDG subgroup detail and rank-order precision that the plateau minimum does not guarantee; the substantive conclusions are not artefacts of sample size choice.

[2]The three metrics are: (1) mean SD of SDG coverage shares, how much the topic balance moves across repeated samples; (2) mean semantic gap, how different research and policy language are within the same SDG; (3) policy–research score gap, how much the scoring instrument favours policy text over research text regardless of content.

Balanced-subset rank stability. The ladder holds the policy side fixed while the research side grows, so the two sides are never size-matched within a tier. Because research (2,536,771 abstracts) outweighs policy (6,367 documents) by about 398:1, this study also tested whether the within-SDG semantic-gap *ranking* survives when the research side is subsampled to a size comparable to the policy corpus. Across 100 independent draws at the 50,000-paper tier, the gap ranking reproduces the full-corpus ordering at Spearman $\rho = 0.986$ (SD 0.009), already $\rho = 0.936$ at 10,000 papers and $\rho = 0.990$ at 100,000 papers, converging to $\rho = 1.000$ at the full corpus.

F Model Selection: Grid Search Protocol and Rationale

Three model families were considered under 5-fold GroupKFold cross-validation (document-level grouping, not random splits) on the pooled training data (n=52,835); the held-out test set (n=9,338) was never touched during any grid search. Logistic regression, which exposes interpretable per-dimension coefficients, and a multi-layer perceptron (MLP) were compared; random forests and gradient boosting were excluded because the embedding space is already a learned representation, so bagged trees add ensemble complexity without a mechanism for improvement or per-dimension interpretability.

The LR grid searched C $\in \{0.1, 1, 2, 3, 5, 10, 30, 100\}$ with an L2 penalty and class_weight $\in$ {None, balanced}; the MLP grid swept n_layers $\in \{2, 3, 4, 6\}$ × hidden_size $\in \{256, 384\}$ × lr $\in \{3e{-}4, 1e{-}3\}$ with weight_decay=0, dropout=0.3, AdamW, and early stopping. Grid search over architecture depth, hidden size, and learning rate revealed negligible performance variance across the MLP configurations (CV macro-F_1 = 0.815–0.819), and LR is flat within ∼0.001 for C $\in \{1, 3, 5\}$, confirming that architecture is not a binding constraint (Table 11).

Table 11. Model selection ranking: 5-fold GroupKFold cross-validated macro-F_1.

Rank	Config	CV macro-F_1
1	MLP 3L/256h (best)	0.8192 ± 0.0032
2	MLP 4L/256h	0.8169 ± 0.0024
3–16	MLP (remaining 14 configurations)	0.815–0.819
17	LR C=3, L2	0.8101 ± 0.0020
18	LR C=1, L2	0.8087 ± 0.0025

Champion: LR (C=3.0, L2). LR (C=3.0, L2, class_weight=None) was chosen over the highest-scoring MLP despite a 0.0091 macro-F_1 gap: LR requires one free parameter (C) and no hidden layers, dropout, learning-rate tuning, or early stopping, and it provides 768 signed coefficients per SDG class, each directly attributable to a specific semantic dimension, whereas MLP's hidden layers diffuse this information across non-linear transformations. For this study's purpose – building an interpretable supervised reference to compare corpora, not a classifier built to label single documents – LR is chosen for transparency, not superior accuracy. The MLP test macro-F_1 (0.826) cited in Section 4.1 is a single held-out evaluation of the champion MLP retrained on the full training pool (n=52,835), distinct from the CV mean of 0.8192 $\pm$ 0.0032 above.

G Register Removal: Convergence and Validation

G.1 Convergence

The full INLP procedure and identification argument are given in Methodology (Section 3.8). This appendix reports the convergence behaviour of the procedure and the construct validation of the component it removes. The adjusted results are the primary decomposition, not sensitivity diagnostics.

Convergence results. Held-out accuracy dropped from 0.978 (iteration 1) to 0.498 by iteration 62, at which point no further linear register separation was detectable. The Spearman rank correlation between the raw (iteration 0, un-projected) and iteration 62 gap vectors was 0.338: the additional removed directions shifted the relative SDG ordering substantially. The mean gap compresses from 0.340 to 0.187, a reduction of 41%, before plateauing. The Spearman ρ versus the raw gap decays from 0.995 (iteration 1) to 0.444 by iteration 15 and then stabilises around 0.352, indicating that the first 15 orthogonal directions capture the bulk of the register signal. Table 12 reports the per-iteration diagnostic underlying this summary across all three encoder tracks.

The decomposition surfaces the SDG 17 inversion discussed in Results (Section 4.2) and Table 2: the register component is positive for all SDGs except SDG 17 (range: +0.155 to +0.261), the only negative value, where register *increases* the apparent research–policy similarity of the partnership goal.

Cross-config replication. The same iterative procedure was run on the MiniLM and SciBERT encoder tracks (the latter two on the S1 subset, by design); all three reach the ≤ 0.5 stopping threshold. Held-out register-classification accuracy falls from 0.965–0.978 at iteration 1 to 0.491–0.498 at convergence, and all three compress sharply in the first 15 iterations (to 0.539–0.685) before plateauing at the majority-class baseline. The iteration count required to reach the threshold differs across encoders, but the qualitative convergence, a rapid early drop followed by a plateau, is identical, confirming that the stopping behaviour reported above is a property of the INLP procedure rather than of the MPNet embedding space.

Table 12. Stratified iterative register-removal diagnostic across the three encoder tracks.

	MPNet			MiniLM			SciBERT		
It.	acc	gap	p vs raw	acc	gap	p vs raw	acc	gap	p vs raw
1	0.978	0.315	0.995	0.965	0.312	0.988	0.973	0.014	0.877
2	0.938	0.303	0.985	0.913	0.285	0.976	0.872	0.011	0.792
3	0.895	0.290	0.983	0.852	0.265	0.951	0.755	0.011	0.748
4	0.864	0.282	0.980	0.796	0.248	0.922	0.691	0.010	0.757
5	0.831	0.279	0.973	0.763	0.234	0.882	0.634	0.010	0.738
10	0.747	0.237	0.708	0.633	0.188	0.532	0.581	0.011	0.667
15	0.685	0.206	0.444	0.584	0.173	0.392	0.539	0.012	0.608
20	0.593	0.195	0.365	0.545	0.167	0.382	0.582	0.014	0.588
30	0.558	0.188	0.390	0.500	0.162	0.336	0.634	0.023	0.515
40	0.535	0.186	0.355	0.498	0.161	0.328	0.596	0.032	0.463
50	0.523	0.187	0.338	—	—	—	0.581	0.034	0.485
60	0.512	0.187	0.338	—	—	—	0.562	0.036	0.483
62	0.498	0.187	0.338	—	—	—	—	—	—
70	—	—	—	—	—	—	0.552	0.038	0.500
79	—	—	—	—	—	—	0.491	0.041	0.471

Notes: acc = held-out accuracy on a stratified 15% test split after projecting out the first k learned directions. gap = average within-SDG semantic gap after k removals. p vs raw = Spearman rank correlation of the gap vector at iteration k versus the raw (iteration 0, un-projected) gap vector; reported per encoder track. Cells marked — indicate an iteration not shown for that track.

G.2 Construct Validation

G.2.1 Purpose and Validation Question

Section 3.8 establishes *what* INLP removes, a corpus-separable linear component, but not what that component is. This subsection evaluates the register interpretation empirically, using an independent surface-linguistic register score computed from the segment texts, following Biber-style register features (Biber, 1988).

The validation can establish three things: whether the adjustment removes a large corpus-separable component; whether the removal is accompanied by evidence that SDG (topic) structure is being destroyed; and whether apparent residual register signals survive controlled samples. It cannot establish that the removed subspace is identical to the measured linguistic register features, and the register score used here is a small set of surface indicators; the validation is single-encoder and sample-limited.

G.2.2 Register Operationalisation and Sample Construction

What the test checks. The main analysis removes a writing-style (register) component with INLP and reads the remaining gap as topic. The appendix checks that reading with independent, surface-level style measurements. Two questions are asked: (1) does removal actually weaken the research–policy distinction? (2) does removal destroy topic? The short answer is yes to (1) and no to (2); whether the removed part is truly *register* is not proven (see the conclusion).

Sample construction. All results use the canonical MPNet track, 408 segments per sample (12 per SDG per corpus, seed 42), and the same frozen projection matrix G as the main analysis. A small set of long flagship reports (the SDSN Sustainable Development Reports 2024 and 2025, the UNDP Human Development Report 2021/22, the WHO report on ethics and governance of AI for health, the EU AI Act, and two UN SDG progress reports) contribute 25 segments across 15 of the 17 SDGs, often as near-duplicates of the same document. These segments are style outliers (mean sentence length 276.9 words vs. 35.6 for the remaining units; nominalisation rate 20.0 vs. 38.8) and, because segments of one document sit close in embedding space, can create fake pooled correlations. The main test therefore uses a one-segment-per-document design (global deduplication by source document; 408 distinct parents, no multi-parent units), which caps each flagship document at one segment and replaces excluded segments with other policy segments from the same SDG, drawn mostly from shorter national and thematic monitoring reports. The two designs differ in makeup, which the Limitations return to. Table 13 summarises both.

Register score. For every sampled segment, six style features are counted per 1,000 words: hedge-word rate, deontic-modality rate (must/should/shall/will), passive-voice frequency, mean sentence length, first-person-pronoun rate, and nominalisation rate; the first principal

Table 13. Validation sample construction: the two designs.

Design	Segments	Research	Policy	Distinct docs	Multi-parent units	Mega-doc segments	Mega-SDGs
Original (per-SDG dedup)	408	204	204	390	25	25	15
One-per-parent (global dedup)	408	204	204	408	0	0	0

Notes: The one-per-parent sample is a rebuild, not a subset of the original sample: excluded flagship segments are replaced by shorter national and thematic monitoring reports, so the two samples differ in composition. Each sample holds 12 segments per SDG per corpus.

component of the standardised features gives the register score. The orientation (positive loading on sentence length) is a sign convention only; the score does not rest on any single feature.

Corpus feature contrasts. Policy segments use markedly more deontic language (3.130 vs. 0.780 per 1,000 words) and research segments hedge more (1.427 vs. 0.854) and passivise more (10.303 vs. 8.453); these contrasts are stable across samples and point in the expected directions for institutional-modality versus academic writing. First-person and nominalisation rates are roughly balanced at the corpus level (6.463 vs. 6.901; 39.003 vs. 36.367), with the flagship segments the outliers on both. Mean sentence length is the one feature whose corpus direction is not robust: the apparent policy lead (63.803 vs. 37.035 words) is driven by the flagship documents and reverses once they are set aside, so sentence length is treated as composition-sensitive rather than load-bearing. Table 14 shows all six features.

Table 14. Corpus register-feature contrasts (original sample).

Feature	Corpus mean		Direction	Load-bearing?	Mega-doc vs rest	
	Policy	Research			Mega-doc	Rest
Hedge rate (/1k words)	0.85	1.43	R > P	Yes	1.21	1.14
Deontic (must/should) rate (/1k words)	3.13	0.78	P > R	Yes	2.21	1.94
Passive-voice rate (/1k words)	8.45	10.30	R > P	Yes	3.35	9.77
Mean sentence length (words)	63.8	37.0	P > R	No	276.9	35.6
First-person rate (/1k words)	6.46	6.90	≈	No	19.00	5.88
Nominalisation rate (/1k words)	39.00	36.37	≈	No	20.02	38.84

Notes: Policy and Research are corpus-mean feature values on the original sample. Mega-doc = mean among the 25 flagship segments; Rest = mean among the remaining units. Rates are per 1,000 words except mean sentence length (words). Load-bearing? marks features used as stable corpus-style signals; sentence length is flagged composition-sensitive.

G.2.3 Corpus Discrimination Before and After INLP

A 5-fold logistic-regression classifier separates research from policy segments almost perfectly on raw embeddings (accuracy 0.909 on the original sample; 0.944 on the one-per-parent sample; Table 15). After register removal the accuracy falls to 0.505 on the original sample (95% CI [0.457, 0.553], binomial $p = 0.441$ against 0.5) and to 0.603 on the one-per-parent sample (95% CI [0.555, 0.649], $p = 1.9e-05$). The adjustment therefore removes a large share of the corpus-linear signal, though not all: the one-per-parent value stays clearly above chance.

A per-SDG-deduplicated variant of the sample (dominated by the 25 flagship policy segments) yields an adjusted accuracy of 0.505, indistinguishable from chance; excluding those segments raises it to 0.574 (difference +0.098, 95% CI [0.025, 0.169]), confirming that flagship clustering drives the original-sample result. The canonical one-per-parent estimate (0.603) is the primary result; residual above-chance separation after removal is not a method failure, since INLP's stopping rule converges on its own training distribution while the validation classifier is re-estimated from scratch.

The register-only rows of Table 15 bound the interpretation. A classifier trained on the six surface features alone reaches only 0.456 on the original sample and 0.544 on the one-per-parent sample, barely above chance in the latter case. The linguistic markers therefore capture only a part of the corpus distinction. This is consistent with the register reading (the corpora do differ in register-like surface ways), but it also delimits it: the removed embedding component is far broader than these six surface features, so the validation cannot claim that the removed subspace is exhausted by, or identical to, the measured features.

G.2.4 Topic Preservation

If the adjustment removed topic, 17-way SDG selectivity, how well the SDG of a segment can be predicted from its embedding, would fall. It does not (Table 16): on the original per-SDG-dedup sample (draw 1), LR cross-validated accuracy is 0.691 on raw and 0.672 on adjusted embeddings, and kNN 0.554 and 0.578 (chance 0.059). The change is small and of mixed sign, so there is no evidence that topic structure is systematically removed; the analysis does not claim perfect preservation, and it was run on the original sample only, not re-estimated on the one-per-parent rebuild. Topic remains linearly decodable after register removal, as the identification argument anticipates.

G.2.5 Residual Register Diagnostics and Robustness Investigation

Two checks examine whether residual register survives adjustment. First, the register score correlates positively with the amount of embedding removed on the per-SDG-deduplicated sample (+0.102, p = 0.040; within-research $\rho = +0.212$, within-policy $\rho = +0.191$). Second, distance to the SDG centroid in adjusted space correlates more strongly with the register score than in raw space (raw $\rho = +0.126$ rising to adjusted $\rho = +0.247$). Both signals are absent on the controlled one-per-parent sample: the removed-norm correlation becomes $\rho = +0.092$ (n.s.; within-corpus -0.043 and -0.036, both n.s.), and the centroid-distance correlation becomes negative (raw $\rho = -0.212$; adjusted $\rho = -0.197$).

The original patterns are therefore attributable to flagship-document clustering rather than to residual register.

A within-SDG decomposition (register score vs. distance to the *other-corpus* SDG centroid, adjusted space) yields one nominally surviving trace among policy segments on the initial draw ($\rho = -0.197$, $p = 0.005$). A draw-stability check re-samples the one-per-parent construction with fresh random streams (seeds 43–45): the statistic is draw-unstable, with the sign flipping across draws and the mean near zero, and flagship exclusion is likewise draw-specific; only 0 of 17 SDGs reach $p < 0.05$ (14 negative, 3 positive). The trace is therefore draw-level noise, not residual register structure.

G.2.6 Validation Conclusion

The validation supports the following claims.

- The adjustment removes a substantial corpus-level component: corpus-discrimination accuracy falls from near-perfect on raw embeddings to moderately above chance after removal, on both the original and one-per-parent samples (Table 15).
- The removed component is not strongly attributable to SDG topic loss: 17-way SDG selectivity is essentially unchanged (Table 16).
- Apparent residual register signals were explained by sampling and document-clustering effects rather than by stable residual structure: the initial removed-magnitude and centroid-distance associations vanished or reversed on the controlled sample, and the one surviving within-SDG trace was draw-unstable.
- The remaining evidence provides partial empirical support for the register reading: the corpora differ in register-like surface features in stable directions, and the adjustment removes corpus-separable signal while leaving topic structure intact, but a measured correspondence between the removed component and the six features was not established, and residual corpus-linear signal remains.

The register interpretation is therefore consistent with the validation evidence rather than confirmed by it; the main text describes it as resting on partial, not complete, empirical support.

G.2.7 Limitations

The validation is bounded in ways that should be read alongside the main text. First, the register score is operationalised as the first principal component of six hand-specified features, and the choice is not resolved here: an alternative a-priori “institutional” z-score combination gave null or reversed results on the initial screen, and the score is oriented by a sign convention rather than by a substantive prior. Second, each sample contains only 408 segments (12 per SDG per corpus), and the per-SDG tests within the draw-stability stage have $n = 12$ policy units each, so the 0-of-17 significant per-SDG result is not evidence of absence at the per-SDG level. Third, the draw-stability check was applied to the within-SDG trace only; the corpus-level accuracies in Table 15 were not re-estimated on fresh draws. Fourth, the one-per-parent sample is a rebuild, not a subset of the original sample, and the excluded flagship documents are replaced by shorter national and thematic monitoring reports, so the two-sample comparison is partly compositional. Fifth, the validation is MPNet-canonical only. Sixth, the residual corpus-linear signal that survives adjustment is not characterised further; its exact linguistic content remains open, part of why the empirical support is partial rather than complete.

Table 15. Corpus discrimination accuracy (research vs. policy, 5-fold logistic regression) before and after INLP register removal.

Sample	Input	n	Fold-mean acc.	Pooled acc. (k/n)	Wilson 95% CI	Binom. p
Original (per-SDG dedup)	register-only	408	0.456	0.456 (186/408)	[0.408, 0.504]	0.967
Original (per-SDG dedup)	raw embeddings	408	0.909	0.909 (371/408)	[0.878, 0.933]	< 0.001
Original (per-SDG dedup)	adjusted (INLP)	408	0.505	0.505 (206/408)	[0.457, 0.553]	0.441
One-per-parent	register-only	408	0.544	0.544 (222/408)	[0.496, 0.592]	0.042
One-per-parent	raw embeddings	408	0.944	0.944 (385/408)	[0.917, 0.962]	< 0.001
One-per-parent	adjusted (INLP)	408	0.603	0.603 (246/408)	[0.555, 0.649]	< 0.001
Original (per-SDG dedup)	adjusted, mega-policy excluded	383	0.574	0.574 (220/383)	[0.524, 0.623]	0.002

Notes: Pooled accuracy = proportion of correct predictions across the five test folds. Wilson 95% CI and one-sided binomial test against 0.5 on the pooled proportion. In the one-per-parent sample each segment comes from a distinct source document, so no test segment shares a source document with its training folds; in the original per-SDG-dedup sample, segments of the same flagship document can straddle folds, so the original-sample rows carry a mild grouped-similarity caveat. Mega-policy excluded = original sample minus the 25 policy segments contributed by the 7 multi-segment documents.

Table 16. 17-way SDG (topic) selectivity before and after register removal (original per-SDG-dedup sample, draw 1).

Input	LR CV acc.	kNN(5) CV acc.	Chance
Raw	0.691	0.554	0.059
Adjusted (INLP)	0.672	0.578	0.059

Notes: Five-fold stratified cross-validated accuracy of logistic regression ($C = 10$) and k-nearest-neighbour classification ($k = 5$) (Cover and Hart, 1967) in predicting the SDG of each segment from raw and adjusted embeddings; chance = 1/17. If the adjustment removed topic, these accuracies would fall.

H Supplementary Cross-Method Data

H.1 Cross-Method Coverage and Semantic Gap Values

Tables 23 and 24 report the raw coverage gap and semantic gap values for each SDG across seven encoder–classifier combinations, with an additional concept-retrieval LR column for comparison against the keyword-retrieved research corpus baseline (Appendix A). While Tables 17 and 22 rank these values to compare cross-method stability, the tables below show the unrounded gap magnitudes, allowing the rank separation to be interpreted in absolute terms. Coverage gap is document-weighted (Assumption A19); semantic gap policy centroids use a 50-segment-per-document cap. The zero-shot nearest-centroid method is reported for the canonical MPNet encoder only (Appendix H.4).

H.2 Rank Robustness: Cross-Sensitivity and Encoder Sensitivity

Tables 17 (adjusted) and 18 (raw) report each SDG's semantic-gap rank under the canonical MPNet–LR configuration and its alternatives: segment cap (20, none) and concept-based retrieval (LR, MLP) in both panels, with policy source family (curated AI/SDG, SDGi, UNGDC, full) added in the raw panel. Tables 19, 20, and 21 report the within-SDG gap rankings under MPNet, MiniLM, and SciBERT for each classifier. The adjusted (register-removed) ranking is reported separately from the raw (naive) ranking, with the tested dimension nested inside each table.

H.3 Encoder Sensitivity

The semantic-gap ranking is moderately preserved across the three encoders on the adjusted (register-removed) gap (MPNet–LR vs. SciBERT–LR Spearman $\rho = 0.54$; MPNet–LR vs. MiniLM–LR $\rho \approx 0.63$), and the coverage-gap ranking is highly stable (MPNet–LR vs. SciBERT–LR $\rho = 0.93$). MiniLM and SciBERT are scored on a deterministic 100,000-paper subset (seed 42) of the canonical segmented corpus, so the architecture comparison isolates encoder choice from corpus scale. Under the raw gap, SDG 3 leads for MPNet–LR and SciBERT–LR while SDG 13 leads for MiniLM–LR; SDG 17 trails in all three (rank 16 or 17). The scientific-domain SciBERT compresses the magnitudes but does not change which goals diverge most. Classifier sensitivity within MPNet on the adjusted gap: the three classifiers, LR, zero-shot nearest-centroid, and the MLP robustness check, show varied agreement (MPNet LR vs. MLP $\rho = 0.85$, LR vs. zero-shot $\rho = 0.60$); under the raw gap, SDG 17 is the most classifier-sensitive goal, sitting at rank 17 under LR but rank 8 under zero-shot. The LR baseline is the canonical reference, with MLP and zero-shot as uncertainty bounds.

H.4 Assignment-Method Comparison: Supervised vs. Nearest-Centroid

The zero-shot nearest-centroid method is used in this study for one purpose only: a single comparison between the canonical supervised classifier (LR, with the MLP as robustness) and an assignment rule with no trained decision boundary. Table 25 reports how often the two classifiers agree, per SDG and overall, for the research corpus (segment level) and the policy corpus (segment level), together with the LR-versus-zero-shot semantic-gap rank delta per SDG.

The comparison confirms that the two classifiers agree on the majority of assignments: the overall LR-versus-zero-shot agreement rate is 67.3% over the 3,105,144 research segments and 81.5% over policy segments (Table 25). Agreement is not uniform across SDGs, and the disagreements are concentrated where the reference centroids are least separated, in line with Appendix B.2. SDG 17, whose reference centroid is the flattest and most isolated in that analysis, is also the most classifier-sensitive goal here: it shows the largest rank delta between the two semantic-gap rankings ($|\Delta| = 9$) and one of the lowest per-SDG classifier agreement rates on the research side (26.1%, second only to SDG 10 at 23.6%). SDG 8, whose supervised rank is also heavily classifier-dependent (rank 6 under LR, rank 14 under zero-shot), shows one of the largest rank deltas ($|\Delta| = 8$, tied with SDG 9). The zero-shot rule, having no supervised decision boundary, anchors partnership and inequality discourse less sharply than the LR classifier. The MLP robustness check agrees with the zero-shot rule at 76.3% of policy segments, close to the LR-versus-zero-shot rate, and at 59.5% of research segments overall.

H.5 Raw-Value Correlation: Metric-Sensitivity Companion

Table 26 repeats the H1a–H1d grid of Table 3 with one change: cells report Pearson r on the raw coverage and gap magnitudes instead of Spearman ρ on their ranks; the grid, predictors, and columns are identical. The four H1 hypotheses are co-equal and each probes a distinct coverage-structure prediction. Under both metrics the adjusted-gap association is stable across the supervised tracks for H1a and H1d (the hypotheses with a detectable signal: positive in 8/9 and 9/9 configs respectively under Pearson) and remains null for H1b and H1c (no consistent signal under either metric – itself informative: coverage divergence tracks the gap and policy share, not dominance or raw research share). The zero-shot config, already scoped as an uncertainty bound, is stable on the adjusted gap. SciBERT is designated a lower-trust robustness encoder (its scientific pretraining compresses gap magnitudes, 0.04 against 0.34, Section H.3).

Possible causes of the SciBERT divergence. Possible causes range from small-n estimation noise to SciBERT's compressed gap magnitudes and lower assignment fidelity; none is established, and the divergence does not threaten the conclusion, which rests on the supervised tracks where the signal is large (+0.70) and stable across metrics. The raw-value specification therefore confirms, rather than threatens, the rank-based result.

H.6 Detailed Policy Profiles by Source Family

The full policy corpus combines a curated AI/SDG sub-corpus, SDGi VNR/VLR reports, and UN General Debate speeches. These are all policy-facing texts, but they are not interchangeable genres. The family split reported in Table 27 therefore asks a narrow question: whether the full-corpus policy profile is broadly stable across source families or whether it is strongly shaped by corpus composition.

The answer is mixed. The broad international-institutional profile is clearly source-family sensitive. The curated AI/SDG family is much more innovation-oriented, with SDG 9 leading its document-weighted profile, while the UNGDC family is more strongly concentrated on climate, partnerships, and governance, consistent with the UNGD topic distributions reported by Jankin, Baturo, and Dasandi (2024), and the SDGi family is comparatively flat, with SDG 11-style institutional review language at its top. At the same time, the broader full-corpus pattern is not driven by one family alone: both the UNGDC speeches and the full corpus are dominated by the climate–partnerships–governance block, and the within-SDG semantic-gap ordering is broadly preserved across families ($\rho = 0.73$–0.97).

The implication is interpretive rather than corrective. The full policy corpus should be read as international institutional SDG discourse, not as a universal profile of all AI sustainability policy. The scope of what the policy-side distribution represents is accordingly narrower, but the main diagnostic purpose remains valid.

Re-running the interaction tests within each policy source family (Table 28) confirms the null is not an artefact of comparing an AI-specific research intersection with a broader policy union: the curated AI/SDG construction reproduces the full-corpus cancellation (Section 4.3), while the SDGi and UNGDC families show positive research-share associations ($\rho = 0.60$, $p = 0.014$; $\rho = 0.42$, $p = 0.115$). The full-corpus row of Table 28 reproduces the canonical interaction estimates (Section 4.3), confirming the replication is faithful.

Table 17. Cross-sensitivity robustness of within-SDG semantic-gap rankings across policy source, segment cap, and retrieval strategy.

SDG		Segment cap		Retrieval	
	Canon	20	None	LR	MLP
1	8	8	8	4	6
2	16	16	17	17	17
3	2	1	2	3	5
4	11	11	11	14	14
5	13	13	13	6	10
6	15	15	15	15	16
7	10	10	9	13	13
8	9	9	7	5	1
9	4	5	3	7	7
10	7	7	10	9	3
11	6	6	5	2	2
12	12	12	12	10	11
13	5	4	4	11	12
14	14	14	14	12	9
15	17	17	16	16	15
16	3	3	6	8	8
17	1	2	1	1	4
Rank Corr (ρ)	1.00	1.00	0.97	0.75	0.68

Adjusted (register-removed, canonical).

Notes: Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap). The Canon column is the canonical MPNet—LR ranking. Segment-cap columns compare cap 20 and no cap. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the Canon column.

Table 18. Cross-sensitivity robustness of within-SDG semantic-gap rankings across policy source, segment cap, and retrieval strategy.

SDG		Policy source				Segment cap		Retrieval	
	Canon	Curated	SDGi	UNGDC	Full	20	None	LR	MLP
1	5	2	5	5	5	5	5	1	1
2	10	9	11	11	10	10	9	11	10
3	1	8	1	1	1	1	2	3	7
4	15	15	12	15	14	15	14	13	17
5	13	13	14	13	13	13	13	6	5
6	12	10	13	12	12	12	12	15	15
7	4	1	6	2	4	4	4	8	11
8	6	5	7	6	6	6	6	4	2
9	9	14	2	10	9	8	8	12	13
10	14	11	15	16	15	14	15	10	8
11	3	4	3	4	3	3	3	2	3
12	11	6	10	8	11	11	11	14	12
13	2	3	4	3	2	2	1	5	4
14	7	7	9	9	7	7	7	9	6
15	16	12	16	14	16	16	16	16	16
16	8	16	8	7	8	9	10	7	9
17	17	17	17	17	17	17	17	17	14
Rank Corr (ρ)	1.00	0.74	0.91	0.96	1.00	1.00	0.99	0.81	0.68

Raw (naive baseline, un-adjusted).

Notes: Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap). The Canon column is the canonical MPNet—LR ranking. Policy-source columns compare the keyword-retrieved research profile against each policy-source family. Segment-cap columns compare cap 20 and no cap. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the Canon column.

Table 19. Domain-encoder sensitivity of within-SDG semantic-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
1	**8**	**11**	**1**	**8**	**7**	**15**	**1**
2	**16**	**16**	**14**	**17**	**17**	**17**	**16**
3	**2**	**1**	**3**	**2**	**1**	**5**	**12**
4	*11*	*14*	*13*	*7*	*8*	*9*	*11*
5	*13*	*10*	*11*	*1*	*3*	*2*	*6*
6	**15**	**15**	**10**	**16**	**15**	**14**	**17**
7	10	12	8	12	12	7	10
8	9	4	12	11	11	13	14
9	*4*	*9*	*16*	*10*	*9*	*11*	*15*
10	7	5	4	4	4	6	2
11	*6*	*6*	*9*	*13*	*13*	*4*	*5*
12	12	7	5	9	10	16	8
13	**5**	**8**	**6**	**6**	**5**	**3**	**4**
14	**14**	**13**	**15**	**15**	**14**	**10**	**7**
15	17	17	17	14	16	12	13
16	**3**	**3**	**7**	**3**	**2**	**1**	**3**
17	*1*	*2*	*2*	*5*	*6*	*8*	*9*
Rank Corr (ρ)	1.00	0.85	0.60	0.63	0.71	0.54	0.36

Adjusted (register-removed, canonical).

Notes: Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section H.3). LR = logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only (scoped to a single supervised-vs-nearest-centroid comparison, Appendix H.4). Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the canonical adjusted MPNet-LR column.

Table 20. Domain-encoder sensitivity of within-SDG semantic-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
1	*5*	*8*	*7*	*17*	*17*	*16*	*1*
2	*10*	*11*	*10*	*14*	*14*	*14*	*14*
3	**1**	**4**	**4**	**2**	**2**	**1**	**3**
4	**15**	**15**	**15**	**15**	**15**	**10**	**12**
5	*13*	*9*	*16*	*4*	*8*	*11*	*15*
6	**12**	**14**	**5**	**13**	**16**	**6**	**8**
7	**4**	**7**	**1**	**5**	**4**	**4**	**4**
8	*6*	*3*	*14*	*10*	*12*	*15*	*11*
9	**9**	**12**	**17**	**8**	**6**	**7**	**13**
10	*14*	*13*	*12*	*3*	*3*	*5*	*7*
11	3	2	2	6	7	3	2
12	*11*	*5*	*9*	*7*	*5*	*13*	*5*
13	**2**	**1**	**6**	**1**	**1**	**2**	**6**
14	*7*	*6*	*3*	*11*	*10*	*8*	*9*
15	*16*	*17*	*11*	*9*	*13*	*9*	*10*
16	*8*	*10*	*13*	*12*	*11*	*12*	*16*
17	**17**	**16**	**8**	**16**	**9**	**17**	**17**
Rank Corr (ρ)	1.00	0.86	0.53	0.40	0.37	0.46	0.62

Raw (naive baseline, un-adjusted).

Notes: Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section H.3). LR = logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only (scoped to a single supervised-vs-nearest-centroid comparison, Appendix H.4). Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the canonical raw MPNet-LR column.

Table 21. Domain-encoder sensitivity of within-SDG coverage-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
1	**7**	**6**	**7**	**7**	**7**	**7**	**6**
2	*15*	*13*	*14*	*11*	*11*	*11*	*9*
3	**1**	**1**	**2**	**1**	**1**	**1**	**1**
4	**4**	**4**	**3**	**5**	**4**	**3**	**3**
5	*8*	*10*	*8*	*12*	*9*	*8*	*8*
6	**9**	**8**	**12**	**8**	**10**	**12**	**10**
7	**16**	**17**	**6**	**16**	**17**	**17**	**17**
8	11	15	9	9	8	9	11
9	**2**	**2**	4	**2**	**2**	**2**	**4**
10	**14**	**9**	**11**	**15**	**14**	**15**	**14**
11	*17*	*16*	*15*	*13*	*13*	*14*	*13*
12	12	14	13	14	15	13	15
13	**5**	**5**	**5**	**6**	**6**	**6**	**5**
14	**10**	**12**	**16**	**10**	**12**	**10**	**12**
15	*13*	*11*	*17*	*17*	*16*	*16*	*16*
16	6	7	10	3	3	5	7
17	**3**	**3**	**1**	**4**	**5**	**4**	**2**
Rank Corr (ρ)	1.00	0.92	0.75	0.89	0.90	0.93	0.90

Notes: Each cell reports the within-SDG coverage-gap rank (1 = largest gap, 17 = smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section H.3). LR = canonical supervised logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only. Coverage gap is document-weighted (Assumption A19) for all methods. Rank Corr (ρ) is the Spearman correlation of each column's SDG coverage-gap ranks against the canonical MPNet-LR column.

Table 22. Cross-sensitivity robustness of within-SDG coverage-gap rankings across policy source and retrieval strategy.

SDG		Policy source			Retrieval	
	Canon	Curated	SDGi	UNGDC	LR	MLP
1	7	17	4	11	6	6
2	15	8	12	12	12	13
3	1	3	1	2	2	2
4	4	4	3	6	7	14
5	8	11	7	17	8	10
6	9	15	9	16	9	8
7	16	7	17	8	17	17
8	11	9	8	9	15	15
9	2	1	2	4	1	1
10	14	16	13	15	14	9
11	17	5	15	7	10	7
12	12	14	10	10	16	16
13	5	6	16	5	5	5
14	10	13	14	13	11	11
15	13	10	11	14	13	12
16	6	12	6	1	4	4
17	3	2	5	3	3	3
Rank Corr (ρ)	1.00	0.40	0.77	0.54	0.87	0.66

Notes: Each cell reports the within-SDG coverage-gap rank (1 = largest gap, 17 = smallest gap). The Canon column is the canonical MPNet—LR ranking. Coverage gap = |research proportion − policy proportion| per SDG, using document-weighted policy proportions (Assumption A19). Policy-source columns compare the keyword-retrieved research profile against each policy-source family. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. The Segment-cap axis is omitted: coverage gap is segment-cap-independent. Rank Corr (ρ) is the Spearman correlation of each column's SDG coverage-gap ranks against the canonical column.

Table 23. Coverage gap values across embedding architectures, classifier variants, and retrieval strategies.

SDG	Coverage gap (%)							
	MPNet			MiniLM		SciBERT		Concept
	LR	MLP	ZS	LR	MLP	LR	MLP	
1	4.1	4.5	3.8	4.1	4.1	5.1	5.6	3.9
2	1.9	1.5	0.7	2.1	2.3	2.6	2.9	2.5
3	21.0	16.8	17.2	22.0	22.4	22.3	23.2	21.0
4	10.8	8.9	11.7	8.9	8.8	12.8	12.3	3.5
5	3.1	2.2	2.6	1.7	2.4	3.9	3.4	3.3
6	2.7	2.6	1.1	2.3	2.3	2.5	2.4	2.9
7	0.2	0.4	4.0	1.0	0.8	0.6	0.1	0.4
8	2.1	1.1	1.9	2.3	2.5	3.7	2.4	2.0
9	15.8	15.0	9.4	12.5	15.1	17.6	11.6	21.6
10	2.0	2.2	1.3	1.1	1.6	1.6	1.5	2.1
11	0.2	0.5	0.3	1.6	1.6	1.8	1.6	2.7
12	2.1	1.1	1.0	1.6	1.6	2.5	1.0	0.9
13	7.6	5.7	8.5	5.1	5.9	5.9	8.6	7.1
14	2.2	1.9	0.2	2.2	2.2	2.8	2.4	2.5
15	2.0	1.9	0.1	0.1	1.0	1.3	0.4	2.1
16	6.1	3.9	1.7	11.2	11.1	9.1	4.8	8.5
17	11.4	12.1	20.2	8.9	8.5	9.1	13.2	11.5

Notes: Coverage gap = |research proportion − policy proportion| per SDG in percentage points, using document-weighted policy proportions (Assumption A19) for all methods.

Table 24. (continued) Semantic gap values across embedding architectures, classifier variants, and retrieval strategies.

SDG	Semantic gap (cosine)							
	MPNet			MiniLM		SciBERT		Concept
	LR	MLP	ZS	LR	MLP	LR	MLP	
1	0.415	0.413	0.407	0.222	0.235	0.032	0.061	0.827
2	0.299	0.330	0.345	0.297	0.306	0.037	0.041	0.428
3	0.494	0.480	0.436	0.495	0.507	0.056	0.056	0.663
4	0.267	0.267	0.265	0.291	0.298	0.044	0.043	0.402
5	0.278	0.380	0.246	0.470	0.360	0.041	0.040	0.575
6	0.292	0.292	0.428	0.298	0.294	0.045	0.046	0.384
7	0.428	0.429	0.525	0.430	0.414	0.052	0.053	0.514
8	0.387	0.507	0.278	0.336	0.317	0.033	0.045	0.625
9	0.314	0.327	0.243	0.350	0.384	0.045	0.041	0.421
10	0.271	0.317	0.305	0.485	0.434	0.047	0.049	0.458
11	0.449	0.510	0.477	0.397	0.371	0.052	0.057	0.693
12	0.296	0.438	0.369	0.391	0.411	0.039	0.052	0.400
13	0.480	0.525	0.423	0.541	0.526	0.053	0.051	0.598
14	0.358	0.431	0.452	0.330	0.334	0.044	0.046	0.504
15	0.239	0.231	0.325	0.345	0.307	0.044	0.045	0.346
16	0.329	0.332	0.294	0.322	0.322	0.040	0.037	0.541
17	0.177	0.267	0.398	0.241	0.355	0.023	0.030	0.312

Notes: Semantic gap $= 1 - \cos(\mathbf{r}_{\mathrm{SDG}}, \mathbf{p}_{\mathrm{SDG}})$ where $\mathbf{r}_{\mathrm{SDG}}$ and $\mathbf{p}_{\mathrm{SDG}}$ are the research and policy centroids; policy centroids use a 50-segment-per-document cap. All values are cosine units. This is a *values* table reporting the raw (naive) semantic gap; the adjusted (register-removed) ranking is shown in the rank-robustness tables (Section H.2).

Table 25. Classifier agreement between the supervised LR classifier and the zero-shot nearest-centroid rule, by SDG and overall, under the canonical MPNet encoder and keyword retrieval.

	Research (segments)			Policy (segments)		Semantic-gap rank		
SDG	n_{LR}	Agree %	Res MLP %	n_{LR}	Agree %	LR	ZS	$\lvert\Delta\rvert$
1 (No Poverty)	13,459	30.9	14.9	1,700	70.8	5	7	2
2 (Zero Hunger)	78,591	77.6	59.8	1,723	94.1	10	10	0
3 (Good Health)	859,355	75.0	79.8	2,934	87.7	1	4	3
4 (Education)	470,002	82.4	82.6	2,055	91.3	15	15	0
5 (Gender Equality)	64,940	56.3	37.0	2,026	88.5	13	16	3
6 (Clean Water)	35,137	85.3	75.2	1,206	94.6	12	5	7
7 (Clean Energy)	119,467	88.3	81.9	1,808	84.1	4	1	3
8 (Decent Work)	65,500	44.4	20.7	2,016	71.3	6	14	8
9 (Innovation)	577,833	50.8	45.1	4,066	81.0	9	17	8
10 (Reduced Ineq.)	30,323	23.6	12.8	1,123	56.9	14	12	2
11 (Sust. Cities)	146,508	77.2	64.0	1,658	88.3	3	2	1
12 (Consumption)	46,199	60.9	38.5	1,187	91.8	11	9	2
13 (Climate)	121,557	40.5	28.4	3,399	84.5	2	6	4
14 (Life Below Water)	51,458	89.6	69.7	1,634	92.3	7	3	4
15 (Life on Land)	78,102	83.5	70.3	1,828	84.1	16	11	5
16 (Peace & Justice)	337,317	55.6	46.1	5,725	65.0	8	13	5
17 (Partnerships)	9,396	26.1	11.7	4,509	84.1	17	8	9
Overall	3,105,144	67.3	59.5	40,597	81.5			

Notes: Agree % is the share of LR-assigned texts that the zero-shot rule also assigns to the same SDG. Semantic-gap ranks are 1 = largest gap on the raw (naive, unadjusted) gap; $|\Delta|$ is the absolute rank difference between the two classifiers. Classifier agreement is computed over all texts (research: each segment; policy: each segment), without the 50-segment-per-document cap, which applies only to centroid building. The zero-shot rule assigns each text to the SDG with the highest cosine similarity to the train-split reference centroids ($\mathbf{s}_k$); the LR rule uses the trained classifier's argmax. MLP agreement against zero-shot is reported for both corpora: research MLP-vs-ZS agreement is computed from the persisted per-shard research scores and shown as the overall Res MLP % (59.5%), alongside the policy MLP-vs-ZS rates.

Table 26. Pearson r on the raw coverage and semantic-gap magnitudes (not ranked).

	Raw gap	Adj. gap	Register
H1a. Coverage gap ↔ Semantic gap			
MPNet LR	+0.102	+0.694**	-0.382
MPNet MLP	-0.069	+0.441†	-0.404
MPNet ZS	-0.171	+0.230	-0.333
MiniLM LR	+0.068	+0.474†	-0.226
MiniLM MLP	+0.333	+0.611**	-0.093
SciBERT LR	+0.129	+0.152	-0.080
SciBERT MLP	-0.108	-0.189	+0.138
Concept LR	+0.030	+0.445†	-0.357
Concept MLP	-0.092	+0.196	-0.310
H1b. Policy–research dominance ↔ Semantic gap			
MPNet LR	+0.274	+0.196	+0.115
MPNet MLP	+0.050	+0.090	-0.014
MPNet ZS	-0.152	-0.227	+0.033
MiniLM LR	+0.293	+0.119	+0.240
MiniLM MLP	+0.353	+0.150	+0.260
SciBERT LR	+0.460†	-0.156	+0.420†
SciBERT MLP	+0.194	-0.358	+0.484*
Concept LR	+0.117	+0.120	+0.021
Concept MLP	-0.060	+0.046	-0.119
H1c. Research coverage ↔ Semantic gap			
MPNet LR	+0.288	+0.464†	-0.055
MPNet MLP	+0.032	+0.308	-0.197
MPNet ZS	-0.191	-0.117	-0.087
MiniLM LR	+0.302	+0.355	+0.101
MiniLM MLP	+0.429†	+0.438†	+0.132
SciBERT LR	+0.419†	+0.208	+0.029
SciBERT MLP	+0.029	-0.234	+0.262
Concept LR	+0.134	+0.262	-0.085
Concept MLP	-0.074	+0.113	-0.205
H1d. Policy coverage ↔ Semantic gap			
MPNet LR	-0.018	+0.501*	-0.359
MPNet MLP	-0.045	+0.418†	-0.361
MPNet ZS	-0.052	+0.258	-0.244
MiniLM LR	-0.041	+0.381	-0.284
MiniLM MLP	+0.064	+0.466*	-0.269
SciBERT LR	-0.138	+0.728**	-0.813***
SciBERT MLP	-0.344	+0.287	-0.492*
Concept LR	+0.009	+0.269	-0.225
Concept MLP	-0.017	+0.130	-0.154

Notes: Same H1a–H1d grid and column layout as Table 3; significance stars use the two-sided Monte Carlo permutation p-value ($^{***}\,p < 0.001$, $^{**}\,p < 0.01$, $^{*}\,p < 0.05$, $^{\dagger}\,p < 0.10$). Identical inputs and configuration grid to Table 3; only the correlation function changes from Spearman ρ (ranks) to Pearson r (raw magnitudes). ZS is MPNet-only per manuscript scope.

Table 27. Per-SDG policy source-family sensitivity: coverage share and semantic gap.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)
1	4.5 (288)	0.42 (1,689)	1.1 (1)	0.44 (357)	5.9 (250)	0.42 (1,191)	1.8 (37)	0.56 (141)
2	4.3 (274)	0.30 (1,508)	0.0 (0)	0.35 (240)	5.8 (246)	0.29 (1,196)	1.4 (28)	0.45 (72)
3	4.7 (298)	0.49 (2,408)	6.4 (6)	0.36 (643)	6.7 (284)	0.55 (1,707)	0.4 (8)	0.70 (58)
4	4.8 (305)	0.27 (2,055)	0.0 (0)	0.21 (274)	7.0 (296)	0.29 (1,718)	0.4 (9)	0.40 (63)
5	5.1 (324)	0.28 (2,005)	0.0 (0)	0.28 (272)	6.6 (278)	0.28 (1,531)	2.2 (46)	0.41 (202)
6	3.7 (238)	0.29 (1,206)	0.0 (0)	0.35 (137)	5.3 (226)	0.29 (1,038)	0.6 (12)	0.42 (31)
7	4.3 (273)	0.43 (1,578)	1.1 (1)	0.51 (358)	6.0 (253)	0.41 (1,147)	0.9 (19)	0.59 (73)
8	4.3 (276)	0.39 (1,886)	0.0 (0)	0.40 (309)	6.5 (275)	0.40 (1,558)	0.0 (1)	0.54 (19)
9	4.3 (274)	0.31 (2,986)	50.0 (47)	0.27 (1,838)	5.1 (215)	0.54 (1,116)	0.6 (12)	0.46 (32)
10	2.9 (182)	0.27 (1,085)	0.0 (0)	0.33 (258)	4.2 (176)	0.28 (774)	0.3 (6)	0.40 (53)
11	5.3 (338)	0.45 (1,658)	0.0 (0)	0.43 (175)	8.0 (336)	0.46 (1,474)	0.1 (2)	0.57 (9)
12	3.5 (226)	0.30 (1,132)	0.0 (0)	0.38 (178)	5.3 (225)	0.30 (947)	0.0 (1)	0.49 (7)
13	11.3 (718)	0.48 (3,084)	7.4 (7)	0.44 (613)	6.0 (252)	0.44 (1,268)	22.4 (459)	0.58 (1,203)
14	3.8 (241)	0.36 (1,293)	0.0 (0)	0.37 (216)	4.5 (189)	0.35 (856)	2.5 (52)	0.49 (221)
15	4.2 (266)	0.24 (1,484)	0.0 (0)	0.29 (299)	5.7 (239)	0.24 (1,108)	1.3 (27)	0.41 (77)
16	17.3 (1,100)	0.33 (4,890)	12.8 (12)	0.20 (1,092)	5.9 (250)	0.38 (1,395)	40.9 (838)	0.51 (2,403)
17	11.7 (746)	0.18 (3,645)	21.3 (20)	0.18 (859)	5.6 (235)	0.16 (1,322)	24.0 (491)	0.26 (1,464)
ρ	1.00	1.00	0.36	0.73	0.61	0.91	0.45	0.97

Notes: For each policy source family, `pol.% (n)` is the document-weighted policy share (%) with the document count in parentheses; `sem. (n)` is the within-SDG research–policy semantic gap (naive, unadjusted) with the capped segment count in parentheses. Read across a row to see whether the gap is stable or driven by one family.

Table 28. Per-family interaction test: coverage predictors vs the within-SDG semantic gap.

Family	n	Research share (ρ, p)	Coverage gap (ρ, p)	Research share excl. SDG 4 (ρ, p)
Full corpus	17	0.45 (0.071)	-0.01 (0.959)	0.61 (0.015)
Curated AI/SDG	17	-0.06 (0.833)	-0.22 (0.386)	0.05 (0.848)
SDGi VNR/VLR	17	0.60 (0.014)	0.04 (0.873)	0.70 (0.003)
UNGDC speeches	15	0.42 (0.115)	0.35 (0.195)	0.61 (0.024)

Notes: Spearman ρ (with two-sided Monte Carlo permutation p, 100,000 resamples, fixed seed) between each coverage predictor and the within-SDG semantic gap across the reliable SDGs (n). Research share is the research-corpus proportion per SDG (family-independent); the coverage gap is $|\text{research} - \text{policy}|$ using the family's document-weighted policy share. The curated AI/SDG family is the symmetric AI/SDG-vs-AI/SDG control matching the research corpus's AI $\cap$ SDG scope.

I Pooled Regression: Coverage Predictors and the Semantic Gap

This appendix pools all 24 encoder–retrieval–method–cap configurations into a single panel ($N = 408$) and regresses the within-SDG semantic gap on coverage predictors using OLS with cluster-robust standard errors by SDG (17 clusters). The dependent variable is the rank of the semantic gap within each configuration (1–17 per config, ties averaged); predictors are expressed in percentage points (0–100). Table 29 reports the full 23-specification grid across core regressions (Panel A), robustness checks (Panel B), interaction terms (Panel C), and alternative functional forms (Panel D).

The adjusted semantic gap is positively associated with the coverage gap ($b = +0.18$, $p = 0.069$; Panel A, column 1); the coefficient is marginally significant at conventional thresholds but is confirmed by the supervised-only subsample ($p = 0.040$), the MPNet-only subsample ($p = 0.003$), and the raw-DV specification in cosine units ($p = 0.007$). The raw gap and the register component show no detectable association with any coverage predictor, consistent with the cancellation pattern reported in Section 4.3. Policy coverage (the share of policy documents assigned to a given SDG) is a stronger predictor of the adjusted gap ($b = +0.41$, $p = 0.003$), capturing how concentrated policy attention is on each goal. The coverage-gap-by-encoder interaction (Panel C) reveals that the effect is strongest for MPNet and reverses for SciBERT, suggesting that domain-specialised pretraining compresses the adjusted semantic gap regardless of coverage structure.

Table 29. Pooled OLS: coverage predictors and the within-SDG semantic gap rank across 24 configurations.

	Core								
	(1) Adj. gap covgap	(2) Adj. gap dominance	(3) Raw gap covgap	(4) Raw gap dominance	(5) Reg. gap covgap	(6) Reg. gap dominance	(7) Adj. gap rescov	(8) Raw gap rescov	(9) Reg. gap rescov
Coverage gap (|research% − policy%|)	+0.2†	–	+0.1	–	-0.0	–	–	–	–
	(0.1)		(0.2)		(0.1)				
Policy coverage	+0.4**	+0.5**	-0.1	+0.1	-0.3†	-0.3	+0.5**	-0.1	-0.3†
	(0.1)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.1)	(0.2)	(0.2)
MiniLM (vs. MPNet)	+0.1	-0.0	+0.0	-0.0	-0.0	+0.0	-0.0	-0.0	+0.0
	(0.8)	(0.8)	(1.2)	(1.1)	(0.9)	(0.9)	(0.8)	(1.1)	(0.9)
SciBERT (vs. MPNet)	-0.1	-0.0	-0.0	-0.0	+0.0	+0.0	-0.0	-0.0	+0.0
	(1.1)	(1.1)	(1.0)	(1.0)	(1.3)	(1.3)	(1.1)	(1.0)	(1.3)
Concept (vs. keyword)	-0.0	+0.0	-0.0	-0.0	+0.0	-0.0	+0.0	+0.0	-0.0
	(0.7)	(0.7)	(0.8)	(0.8)	(1.0)	(1.0)	(0.7)	(0.8)	(1.0)
Cap 20 (vs. 50)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
	(0.1)	(0.1)	(0.1)	(0.1)	(0.2)	(0.2)	(0.1)	(0.1)	(0.2)
No cap (vs. 50)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	+0.0
	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)
MLP (vs. LR)	-0.0	-0.0	-0.0	+0.0	-0.0	-0.0	+0.0	+0.0	+0.0
	(0.4)	(0.4)	(0.5)	(0.5)	(0.3)	(0.3)	(0.4)	(0.5)	(0.3)
ZS (vs. LR)	+0.0	+0.0	+0.0	-0.0	+0.0	-0.0	+0.0	+0.0	-0.0
	(0.7)	(0.7)	(1.3)	(1.3)	(0.8)	(0.8)	(0.7)	(1.3)	(0.8)
Domination (research% − policy%)	–	+0.1	–	+0.1	–	+0.0	–	–	–
		(0.1)		(0.1)		(0.1)			
Research coverage (%)	–	–	–	–	–	–	+0.1	+0.1	+0.0
							(0.1)	(0.1)	(0.1)
sdg1	–	–	–	–	–	–	–	–	–
sdg2	–	–	–	–	–	–	–	–	–
sdg3	–	–	–	–	–	–	–	–	–
sdg4	–	–	–	–	–	–	–	–	–
sdg5	–	–	–	–	–	–	–	–	–
sdg6	–	–	–	–	–	–	–	–	–
sdg7	–	–	–	–	–	–	–	–	–
sdg8	–	–	–	–	–	–	–	–	–
sdg9	–	–	–	–	–	–	–	–	–
sdg10	–	–	–	–	–	–	–	–	–
sdg11	–	–	–	–	–	–	–	–	–
sdg12	–	–	–	–	–	–	–	–	–
sdg13	–	–	–	–	–	–	–	–	–
sdg14	–	–	–	–	–	–	–	–	–
sdg15	–	–	–	–	–	–	–	–	–
sdg16	–	–	–	–	–	–	–	–	–
covgap$\times i_{\text{minilm}}$	–	–	–	–	–	–	–	–	–
covgap$\times i_{\text{scibert}}$	–	–	–	–	–	–	–	–	–
covgap$\times i_{\text{concept}}$	–	–	–	–	–	–	–	–	–
covgap$\times i_{\text{mlp}}$	–	–	–	–	–	–	–	–	–
covgap$\times i_{\text{zs}}$	–	–	–	–	–	–	–	–	–
N	408	408	408	408	408	408	408	408	408
R^2	0.195	0.162	0.014	0.049	0.073	0.074	0.162	0.049	0.074
Adj. R^2	0.177	0.143	-0.008	0.028	0.052	0.053	0.143	0.028	0.053

	Robustness						Interactions				Functional form			
	(10) No classifier ind.	(11) + SDG FE	(12) Supervised only	(13) Keyword only	(14) MPNet only	(15) No clf + SDG FE	(16) + covgap×enc	(17) + covgap×ret	(18) + covgap×mth	(19) + all interactions	(20) Raw DV (not ranked)	(21) Log DV	(22) WLS	(23) Rank + bootstrap
Coverage gap (|research% − policy%|)	+0.2†	-0.0	+0.2*	+0.2	+0.3**	-0.0	+0.3**	+0.2	+0.2**	+0.4***	+0.0**	+0.0**	+0.4***	+0.2†
	(0.1)	(0.2)	(0.1)	(0.1)	(0.1)	(0.2)	(0.1)	(0.1)	(0.1)	(0.1)	(0.0)	(0.0)	(0.1)	(0.1)
Policy coverage	+0.4**	+1.2†	+0.4**	+0.4**	+0.4*	+1.2†	+0.4**	+0.4**	+0.4**	+0.4**	+0.0*	+0.0*	+0.5***	+0.4**
	(0.1)	(0.7)	(0.1)	(0.1)	(0.1)	(0.7)	(0.1)	(0.1)	(0.1)	(0.1)	(0.0)	(0.0)	(0.1)	(0.1)
MiniLM (vs. MPNet)	+0.1	-0.0	+0.1	+0.1	–	-0.0	+0.4	+0.1	+0.1	+0.5	-0.0**	-0.0**	+0.2	+0.1
	(0.8)	(0.8)	(0.9)	(0.8)		(0.8)	(1.1)	(0.8)	(0.8)	(1.1)	(0.0)	(0.0)	(0.9)	(0.8)
SciBERT (vs. MPNet)	-0.1	+0.0	-0.1	-0.1	–	+0.0	+2.6*	-0.1	-0.1	+2.8*	-0.1***	-0.1***	-3.4	-0.1
	(1.1)	(1.1)	(1.3)	(1.1)		(1.1)	(1.2)	(1.1)	(1.1)	(1.2)	(0.0)	(0.0)	(2.1)	(1.1)
Concept (vs. keyword)	-0.0	+0.0	-0.0	–	-0.1	+0.0	-0.1	-0.6	-0.0	+0.8	+0.1***	+0.1***	-1.0	-0.0
	(0.6)	(0.7)	(0.7)		(0.8)	(0.7)	(0.7)	(1.0)	(0.7)	(0.8)	(0.0)	(0.0)	(1.2)	(0.7)
Cap 20 (vs. 50)	+0.0	-0.0	+0.0	-0.0	+0.0	-0.0	+0.0	-0.0	-0.0	-0.0	+0.0*	+0.0*	-0.1	-0.0
	(0.2)	(0.1)	(0.1)	(0.1)	(0.1)	(0.2)	(0.1)	(0.1)	(0.1)	(0.1)	(0.0)	(0.0)	(0.3)	(0.1)
No cap (vs. 50)	+0.0	-0.0	+0.0	-0.0	+0.0	-0.0	-0.0	+0.0	-0.0	+0.0	+0.0	+0.0	-0.2	-0.0
	(0.3)	(0.2)	(0.2)	(0.3)	(0.2)	(0.3)	(0.2)	(0.2)	(0.2)	(0.2)	(0.0)	(0.0)	(0.4)	(0.2)
MLP (vs. LR)	–	-0.0	+0.0	-0.0	+0.0	–	-0.0	-0.0	+0.6	+0.5	+0.0*	+0.0*	-0.8	-0.0
		(0.4)	(0.4)	(0.4)	(0.6)		(0.4)	(0.4)	(0.5)	(0.5)	(0.0)	(0.0)	(0.9)	(0.4)
ZS (vs. LR)	–	-0.0	–	+0.0	+0.0	–	-0.0	-0.0	+0.8	+0.7	+0.0	+0.0	-0.8	+0.0
		(0.7)		(0.7)	(1.0)		(0.7)	(0.7)	(0.9)	(0.9)	(0.0)	(0.0)	(1.3)	(0.7)
Domination (research% − policy%)	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Research coverage (%)	–	–	–	–	–	–	–	–	–	–	–	–	–	–
sdg1	–	+7.5†	–	–	–	+7.5†	–	–	–	–	–	–	–	–
		(3.9)				(3.9)								
sdg2	–	-2.8	–	–	–	-2.8	–	–	–	–	–	–	–	–
		(4.3)				(4.3)								
sdg3	–	+7.3	–	–	–	+7.3	–	–	–	–	–	–	–	–
		(4.5)				(4.5)								
sdg4	–	+1.5	–	–	–	+1.5	–	–	–	–	–	–	–	–
		(3.6)				(3.6)								
sdg5	–	+4.8	–	–	–	+4.8	–	–	–	–	–	–	–	–
		(3.7)				(3.7)								
sdg6	–	-1.0	–	–	–	-1.0	–	–	–	–	–	–	–	–
		(4.6)				(4.6)								
sdg7	–	+1.6	–	–	–	+1.6	–	–	–	–	–	–	–	–
		(4.4)				(4.4)								
sdg8	–	+2.0	–	–	–	+2.0	–	–	–	–	–	–	–	–
		(4.2)				(4.2)								
sdg9	–	+1.5	–	–	–	+1.5	–	–	–	–	–	–	–	–
		(4.1)				(4.1)								
sdg10	–	+10.1†	–	–	–	+10.1†	–	–	–	–	–	–	–	–
		(5.2)				(5.2)								
sdg11	–	+1.9	–	–	–	+1.9	–	–	–	–	–	–	–	–
		(3.6)				(3.6)								
sdg12	–	+3.5	–	–	–	+3.5	–	–	–	–	–	–	–	–
		(4.7)				(4.7)								
sdg13	–	-1.6†	–	–	–	-1.6†	–	–	–	–	–	–	–	–
		(0.8)				(0.8)								
sdg14	–	+0.8	–	–	–	+0.8	–	–	–	–	–	–	–	–
		(4.6)				(4.6)								
sdg15	–	-2.8	–	–	–	-2.8	–	–	–	–	–	–	–	–
		(4.4)				(4.4)								
sdg16	–	-9.4	–	–	–	-9.4	–	–	–	–	–	–	–	–
		(5.7)				(5.7)								
covgap$\times i_{\text{minilm}}$	–	–	–	–	–	–	-0.0	–	–	-0.1	–	–	–	–
							(0.1)			(0.1)				
covgap$\times i_{\text{scibert}}$	–	–	–	–	–	–	-0.5***	–	–	-0.5***	–	–	–	–
							(0.1)			(0.1)				
covgap$\times i_{\text{concept}}$	–	–	–	–	–	–	–	+0.1	–	-0.1†	–	–	–	–
								(0.1)		(0.1)				
covgap$\times i_{\text{mlp}}$	–	–	–	–	–	–	–	–	-0.1†	-0.1†	–	–	–	–
									(0.0)	(0.0)				
covgap$\times i_{\text{zs}}$	–	–	–	–	–	–	–	–	-0.1	-0.1	–	–	–	–
									(0.1)	(0.1)				
N	408	408	357	357	170	408	408	408	408	408	408	408	408	408
R^2	0.195	0.617	0.198	0.196	0.331	0.617	0.255	0.197	0.200	0.260	0.684	0.701	0.521	0.195
Adj. R^2	0.181	0.592	0.179	0.177	0.302	0.594	0.235	0.177	0.178	0.233	0.677	0.694	0.510	0.177

J Declaration of AI Use

In accordance with the University's guidance on generative AI ("Use of GenAI is permitted to support assignments"), this appendix declares the use of AI tools throughout the dissertation. All use was conducted with human oversight and control; AI-generated outputs were reviewed, modified, and independently verified before inclusion. No AI-generated text was submitted without such verification. This declaration follows the order of permitted activities specified in the assessment guidelines. AI-assisted activities not explicitly listed (e.g., code debugging, plotting, LaTeX troubleshooting, literature summarisation) fall under the same oversight and verification standards described in the relevant rows above.

Table 30. Declaration of AI use by permitted activity.

Activity	**AI tool(s)**	**How output was used**
Researching and refining ideas	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	ChatGPT was used iteratively to develop ideas that aligned with the author's intent. Coding agents independently checked data and code availability to confirm feasibility. All final decisions and conclusions are the author's own. Final research design was formally presented to and approved by supervisor.
Information retrieval / background research	Gemini, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	AI tools read the manuscript for context and recommended papers to investigate further. All cited sources were independently verified against the original publications.
Drafting an outline / organising thoughts	MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Generated code scaffolding and assisted with the full analysis pipeline (embedding, scoring, validation, robustness, appendix outputs). The centroid-assignment step was verified against hand-labeled samples, and the scoring and coverage-gap functions were unit-tested. Analytical decisions were directed by the author.
Refining research questions	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Overlaps with row 1. ChatGPT sharpened framing based on reviewed literature; coding agents confirmed questions were answerable with available data and models. The author retained full control over the final formulation. RQs presented to and approved by supervisor.
Checking spelling and grammar	ChatGPT, Gemini, Claude (web/mobile), MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Coding agents provided inline stylistic and phrasing suggestions during drafting. Web and mobile versions of ChatGPT, Gemini, and Claude were used for editorial review and proofreading. All suggestions were accepted or rejected per standard editorial judgment.